Austo Automobile
PROJECT REPORT

PGPDSGA

Submitted By

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Learning for Life

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1. Context

Austo Motor Company is a leading car manufacturer specializing in SUV, Sedan, and Hatchback models. In its recent board meeting, concerns were raised by the members on the efficiency of the marketing campaign currently being used. The board decides to rope in analytics professional to improve the existing campaign.

2. Objective

Austo Motor Company wants to analyze the data to get a fair idea about the demand of customers which will help them in enhancing their customer experience. Assuming as a Data Scientist at the company, perform the data analysis to find answers to the questions that will help the company to improve the business.

3. Problem Statement

Austo Motor Company's recent marketing efforts have yielded sub-optimal returns, necessitating a data-driven overhaul of their outreach strategy. The objective of this project is to analyze customer purchasing behavior to identify high-value segments and key drivers of demand. By examining the interplay between buyer demographics—including income, profession, and family structure—and product preferences (make and price point), we aim to deliver actionable insights that optimize marketing spend and elevate the overall customer experience.

Austo Motor Company seeks to reverse declining marketing efficiency by leveraging data to decode customer demand. This project identifies the "who, what, and why" of car purchasing—mapping demographic profiles (salary, family, and education) to specific vehicle preferences. Our goal is to transform these insights into a targeted strategy that boosts sales and strengthens brand loyalty.

The company is looking for answers for following:

1. Do men tend to prefer SUVs more compared to women?
2. What is the likelihood of a salaried person buying a Sedan?
3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?
4. How does the amount spent on purchasing automobiles vary by gender?
5. How much money was spent on purchasing automobiles by individuals who took a personal loan?
6. How does having a working partner influence the purchase of higher-priced cars?

4. Approach Followed

1.Get the data, analyze, clean and get the quality of data correct.
2. Find data types are correct or wrong, find missing values and impute as required.
3. Do univariate analysis, bivariate analysis and find out the relation between variables.
4. Gather insights and suggest to company.

5. Data description

• Age: The age of individual in years.
• Gender: The gender of the individual, categorized as male or female.
• Profession: The occupation or profession of the individual.
• Marital_status: The marital status of the individual, such as married &, single
• Education: The educational qualification of the individual Graduate and Post Graduate
• No_of_Dependents: The number of dependents (e.g., children, elderly parents) that the individual supports financially.
• Personal_loan: A binary variable indicating whether the individual has taken a personal loan "Yes" or "No"
• House_loan: A binary variable indicating whether the individual has taken a housing loan "Yes" or "No"
• Partner_working: A binary variable indicating whether the individual's partner is employed "Yes" or "No"
• Salary: The individual's salary or income.
• Partner_salary: The salary or income of the individual's partner, if applicable.
• Total_salary: The total combined salary of the individual and their partner (if applicable).
• Price: The price of a product or service.
• Make: The type of automobile

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Age	1581.0	NaN	NaN	NaN	31.922201	8.425978	22.0	25.0	29.0	38.0	54.0
Gender	1528	4	Male	1199	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Profession	1581	2	Salaried	896	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Marital_status	1581	2	Married	1443	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Education	1581	2	Post Graduate	985	NaN	NaN	NaN	NaN	NaN	NaN	NaN
No_of_Dependents	1581.0	NaN	NaN	NaN	2.457938	0.943483	0.0	2.0	2.0	3.0	4.0
Personal_loan	1581	2	Yes	792	NaN	NaN	NaN	NaN	NaN	NaN	NaN
House_loan	1581	2	No	1054	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Partner_working	1581	2	Yes	868	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Salary	1581.0	NaN	NaN	NaN	60392.220114	14674.825044	30000.0	51900.0	59500.0	71800.0	99300.0
Partner_salary	1475.0	NaN	NaN	NaN	20225.559322	19573.149277	0.0	0.0	25600.0	38300.0	80500.0
Total_salary	1581.0	NaN	NaN	NaN	79625.996205	25545.857768	30000.0	60500.0	78000.0	95900.0	171000.0
Price	1581.0	NaN	NaN	NaN	35597.72296	13633.636545	18000.0	25000.0	31000.0	47000.0	70000.0
Make	1581	3	Sedan	702	NaN	NaN	NaN	NaN	NaN	NaN	NaN

We can see the **five point summary** from here
Minimum value(min), 25% Q1 quantile (25%), 50% Median (50%), 75% Q3 quantile (75%) and maximum value(max) of each column from the above table.

Also, we can find mean, standard deviation, top frequency data in each column.

6. Data Overview & Structure

6.1. Understanding Missing Values

There are 1581 rows and 14 columns.

- Gender has 1528 non-null values, which is less than 1581 rows. This means there are missing values in Gender.
- Partner_Salary has 1475 non-null values, which means it has missing values.
- Numerical columns - Age, No of dependants, Salary, Partner_salary, Total_salary, Price. All others are categorical columns

6.1.1. Gender - Missing Values Analysis

1. Solving the observed inconsistencies like 'Femal' and 'Femle'. Changed to 'Female' in the consistent format.
   Count of Female changed from 327 to 329, which corrected the misspelled ones.
2. Filling the null values with 'NOT Known' value.
   The mode of Gender is 'Male' and then replaced the null values with the 'NOT Known' value. Considered replacing it with the Mode of column, but this approach can give bias in later analysis. Hence progressed with this approach. The total non-null count for Gender column is 1581 which is the same as the number of rows.

Now the Gender column is fine.

6.1.2. Partner_Salary - Missing Values Analysis

Number of missing values in 'Partner_salary': 106

Number of missing 'Partner_salary' values where 'Partner_working' is 'No': 90

Number of missing 'Partner_salary' values where 'Partner_working' is 'Yes': 16

1. Case 1: `Partner_working` is 'No' and `Partner_salary` is missing, we will fill `Partner_salary` with 0.
   Updated the value as 0 in 90 rows.
2. Case 2: `Partner_working` is 'Yes' and `Partner_salary` is missing, we will fill `Partner_salary` with the median salary of working partners. The median is chosen to be robust, because we can see potential outliers, which is suggested by the difference between mean and median.
   Updated the median value of 32900.0 in 16 columns.

6.2. Partner_Salary - Data Type

The salary, total salary attributes are of data type int. Hence partner_salary also can be int, assuming the total salary is the salary of individual and partner together.

7. Univariate Analysis

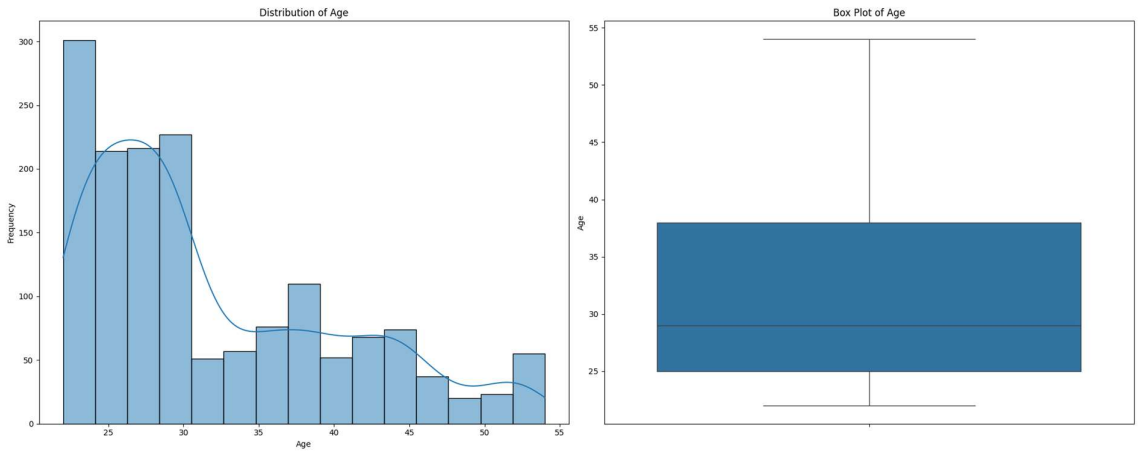
7.1. Univariate Analysis for Numerical:

Let us analyze each of the variable.

7.1.1. Descriptive statistics for Age:

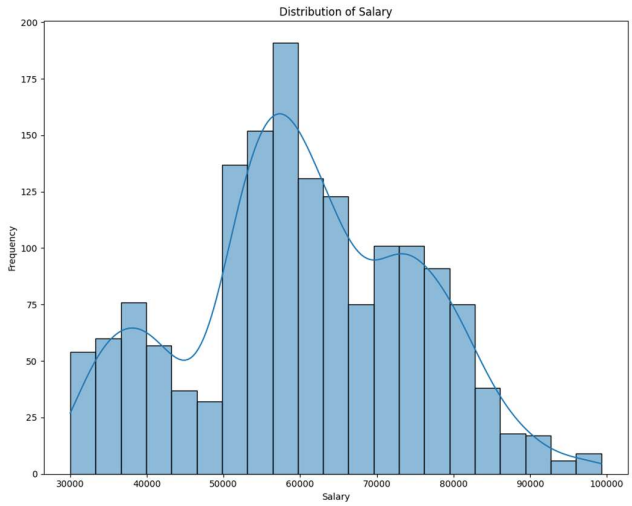
```
count    1581.000000
```

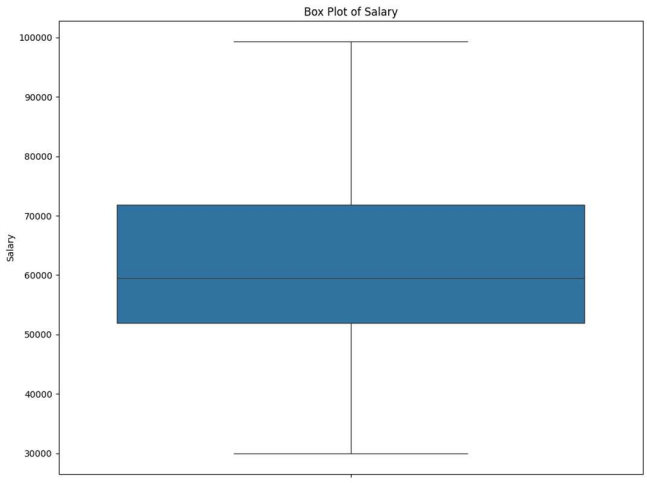
mean	31.922201
std	8.425978
min	22.000000
25%	25.000000
50%	29.000000
75%	38.000000
max	54.000000
Name: Age, dtype: float64	



7.1.2. Descriptive statistics for Salary:

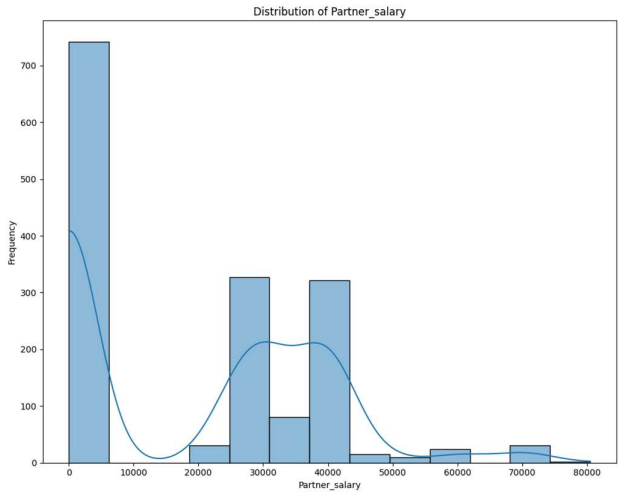
count	1581.000000
mean	60392.220114
std	14674.825044
min	30000.000000
25%	51900.000000
50%	59500.000000
75%	71800.000000
max	99300.000000
Name: Salary, dtype: float64	

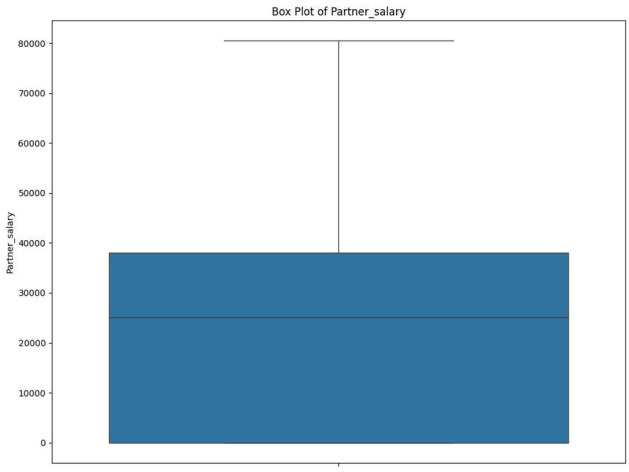




7.1.3. Descriptive statistics for Partner_salary:

count	1581.000000
mean	19202.466793
std	19526.571322
min	0.000000
25%	0.000000
50%	25100.000000
75%	38000.000000
max	80500.000000
Name: Partner_salary, dtype: float64	

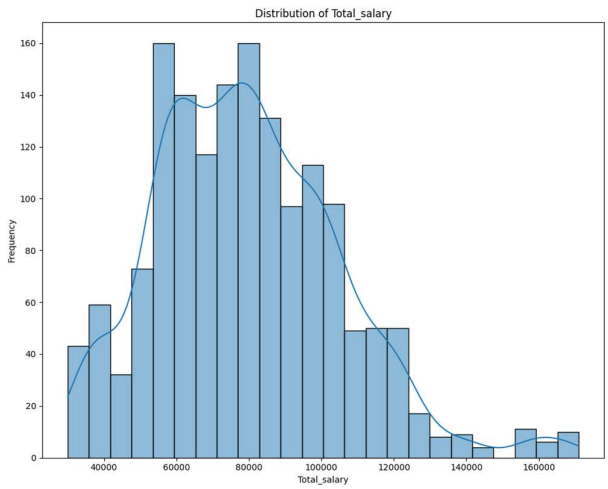


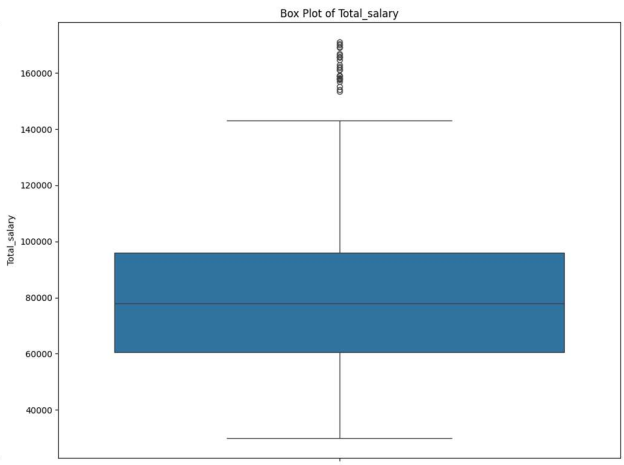


7.1.4. Descriptive statistics for Total_salary:

count	1581.000000
mean	79625.996205
std	25545.857768

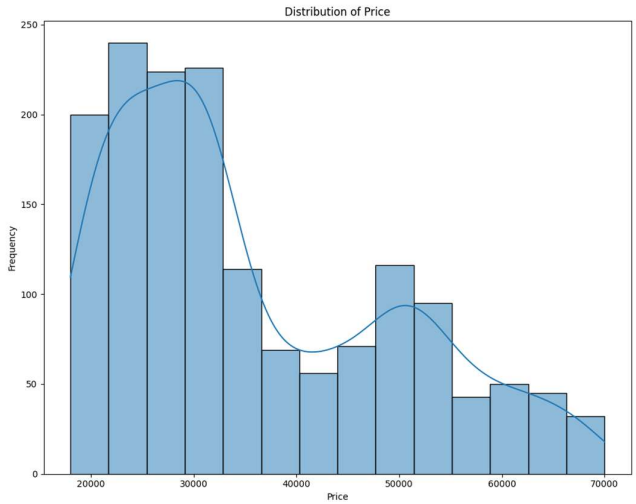
min	30000.000000
25%	60500.000000
50%	78000.000000
75%	95900.000000
max	171000.000000
Name: Total_salary, dtype: float64	

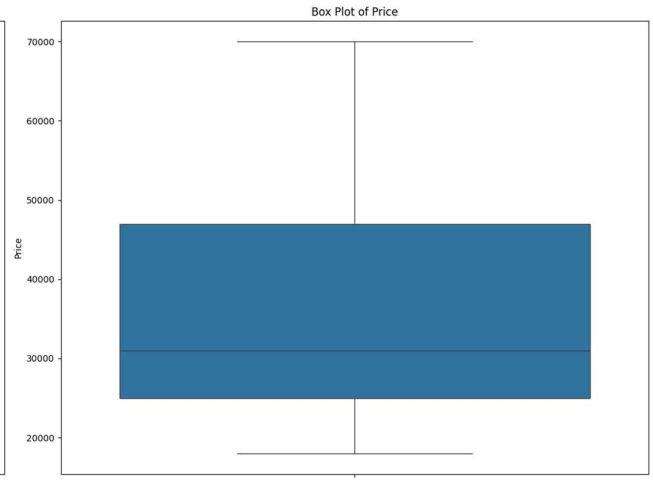




7.1.5. Descriptive statistics for Price:

count	1581.000000
mean	35597.722960
std	13633.636545
min	18000.000000
25%	25000.000000
50%	31000.000000
75%	47000.000000
max	70000.000000
Name: Price, dtype: float64	





7.2. Univariate Analysis for Categorical Columns

1.Identify list of categorical columns ('Gender', 'Profession', 'Marital_status', 'Education', 'Personal_loan', 'House_loan', 'Partner_working', 'Make')

2.Generate bar plots to visualize the frequency distribution of each category and display their value counts.

Steps: To perform univariate analysis on the categorical columns:

1.Define a list of these columns.

2.Iterate through this list, displaying value counts

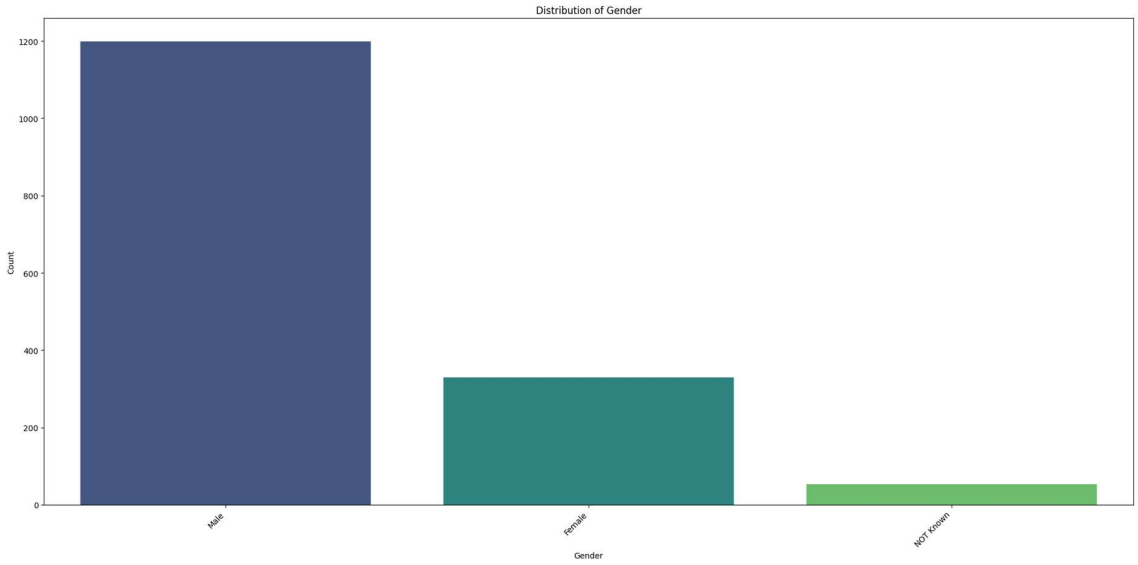
3.Generate a bar plot for each column to visualize its frequency distribution.

7.2.1. Analyzing Categorical Column: Gender

Value Counts for Gender:
Gender

Male	1199
Female	329
NOT Known	53

Name: count, dtype: int64

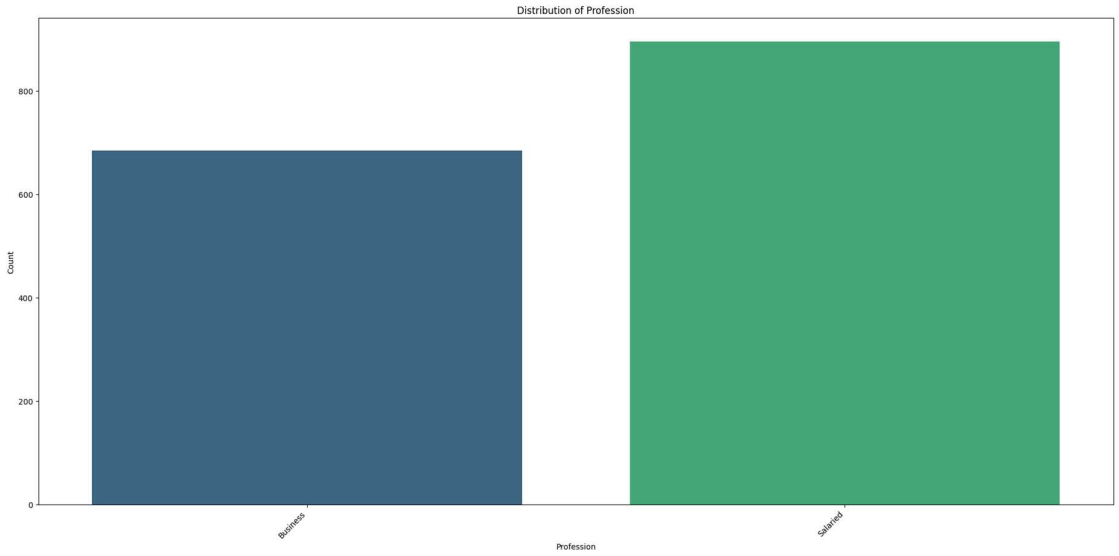


7.2.2. Analyzing Categorical Column: Profession

Value Counts for Profession:
Profession

Salaried	896
Business	685

Name: count, dtype: int64

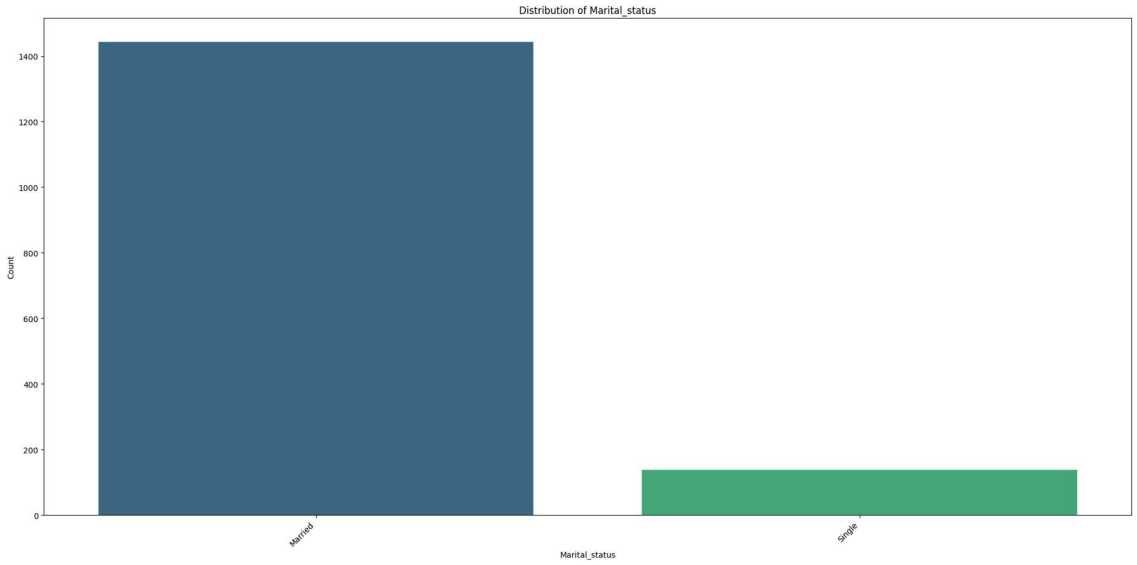


7.2.3. Analyzing Categorical Column: Marital_status

Value Counts for Marital_status:
Marital_status

Married	1443
Single	138

Name: count, dtype: int64

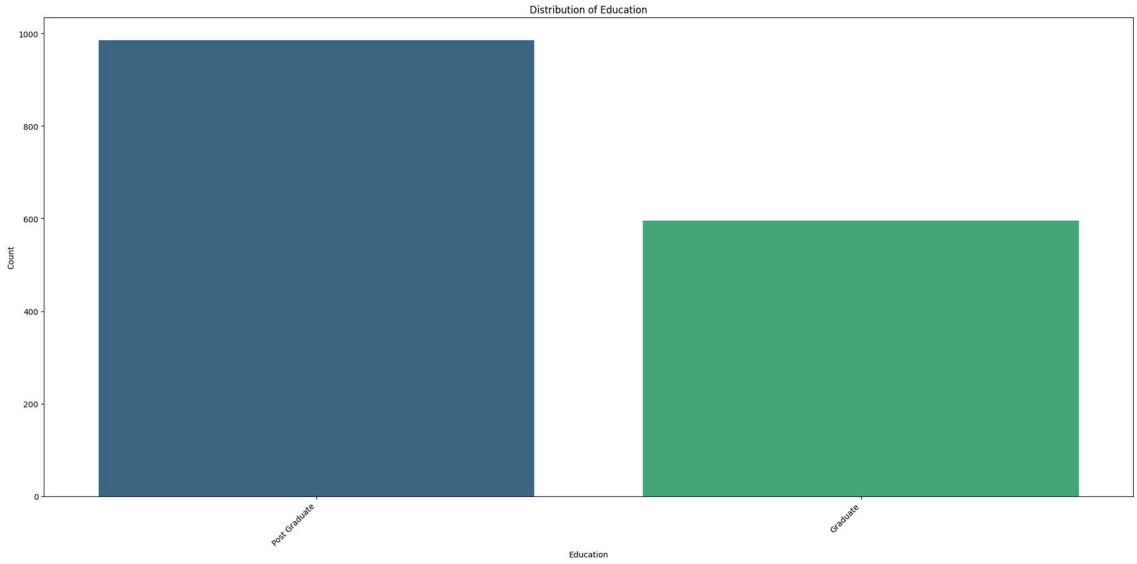


7.2.4. Analyzing Categorical Column: Education

Value Counts for Education:
Education

Post Graduate	985
Graduate	596

Name: count, dtype: int64

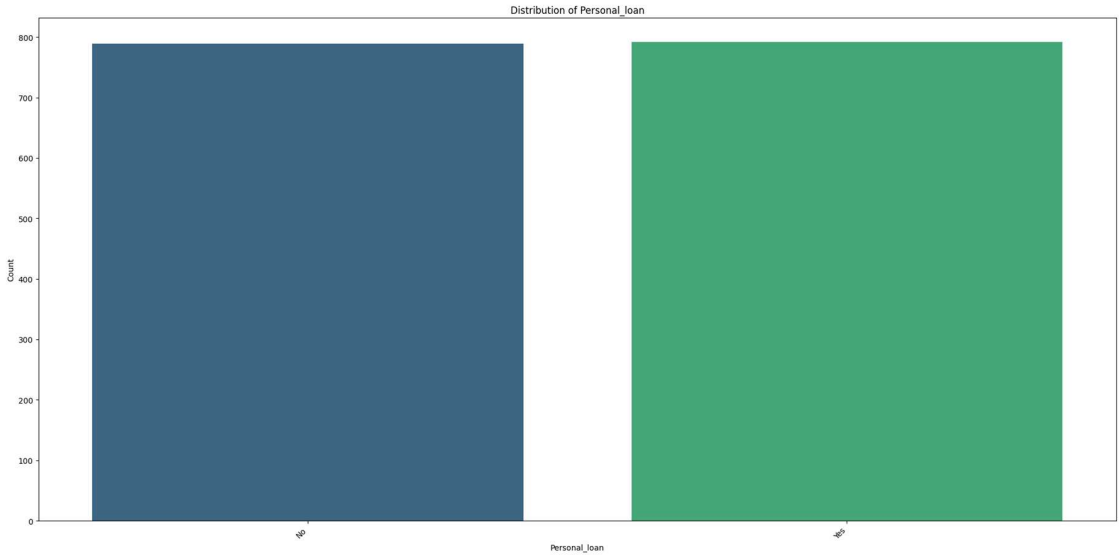


7.2.5. Analyzing Categorical Column: Personal_loan

Value Counts for Personal_loan:
Personal_loan

Yes	792
No	789

Name: count, dtype: int64

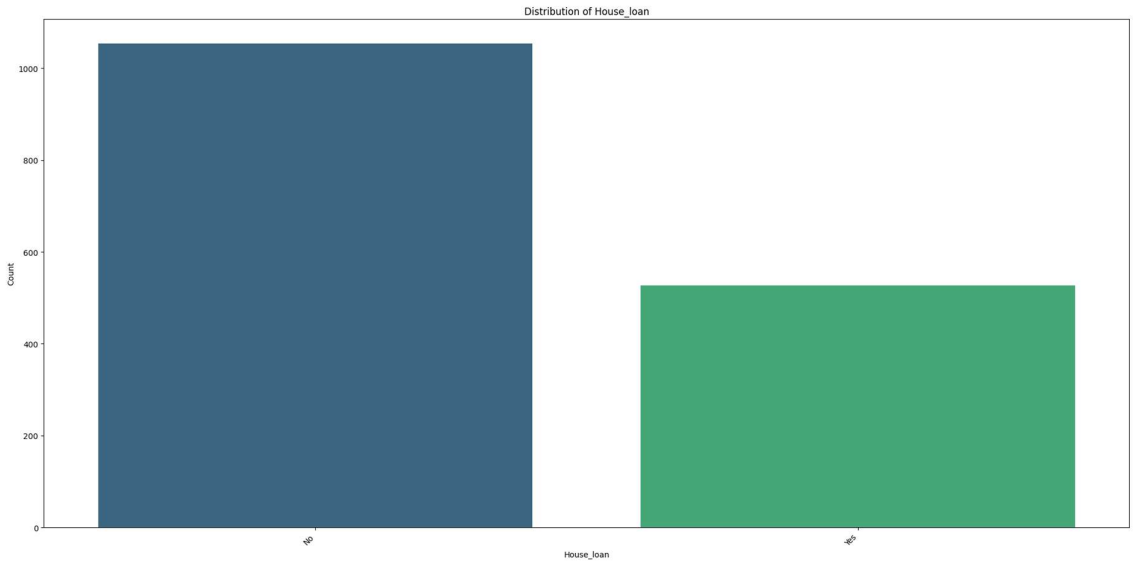


7.2.6. Analyzing Categorical Column: House_loan

Value Counts for House_loan:
House_loan

No	1054
Yes	527

Name: count, dtype: int64

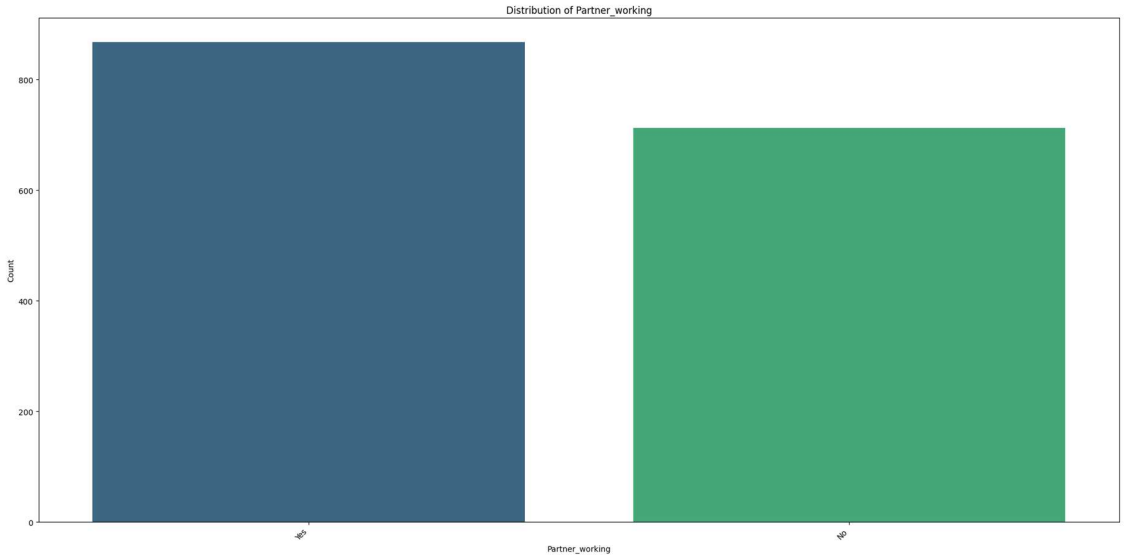


7.2.7. Analyzing Categorical Column: Partner_working

Value Counts for Partner_working:
Partner_working

Yes	868
No	713

Name: count, dtype: int64

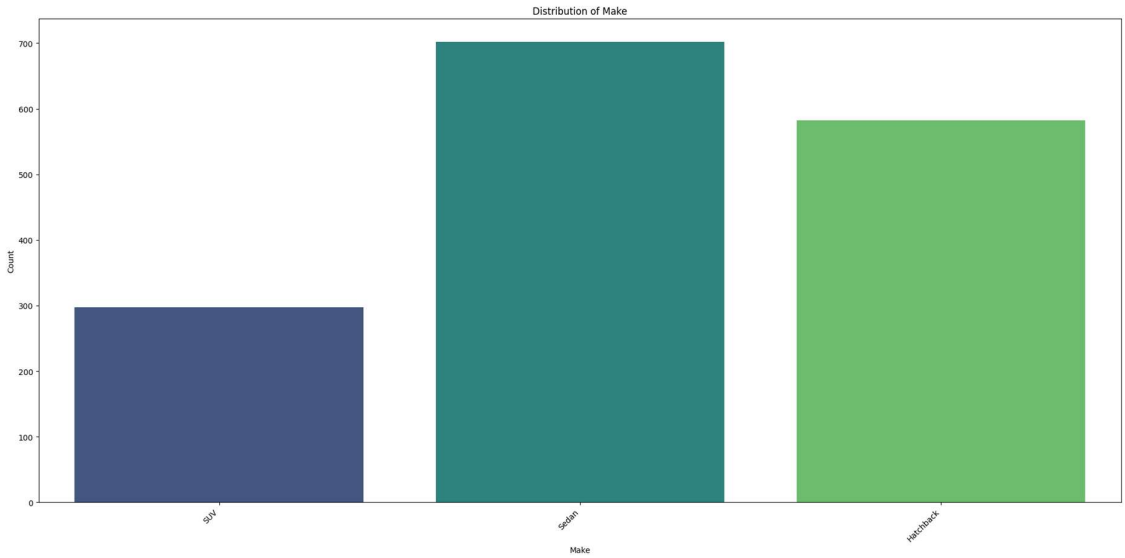


7.2.8. Analyzing Categorical Column: Make

Value Counts for Make:
Make

Sedan	702
Hatchback	582
SUV	297

Name: count, dtype: int64

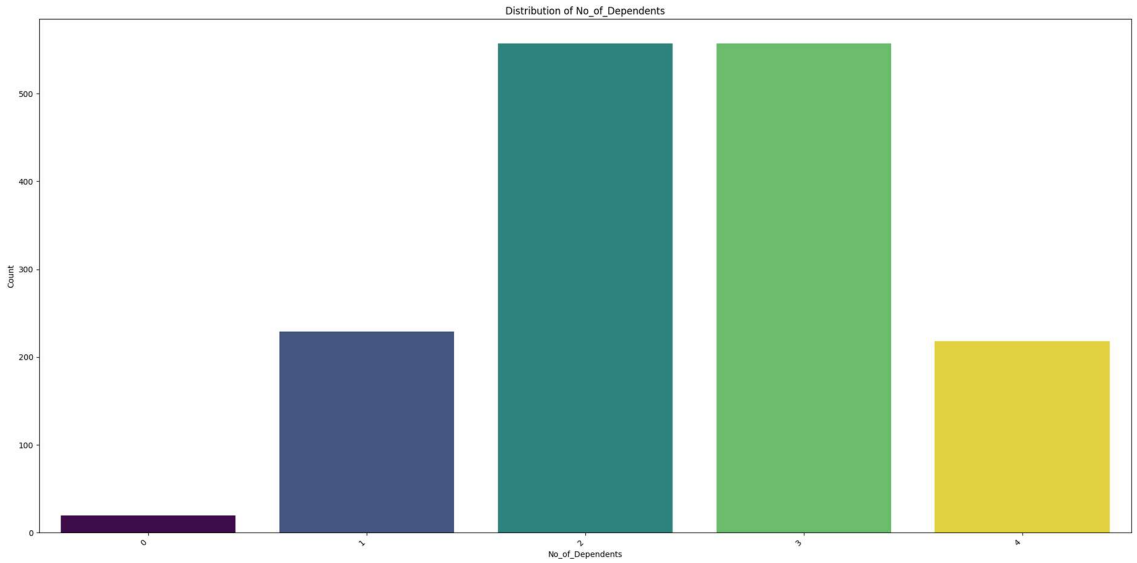


7.2.9. Analyzing Categorical Column: No_of_Dependents

Value Counts for No_of_Dependents:
No_of_Dependents

3	557
2	557
1	229
4	218
0	20

Name: count, dtype: int64



7.3. Summary of Numerical Column Insights (Outliers, Distribution)

Based on the descriptive statistics, histograms, and box plots for the numerical columns, here are the consolidated observations:

Age

- Distribution Shape: The histogram shows a right-skewed distribution, indicating a higher concentration of younger individuals. The peak is around 25-30 years.
- Central Tendency: The mean age is approximately 31.92 years, and the median is 29.0 years, consistent with the right-skewness.
- Spread: Ages range from 22 to 54 years. The interquartile range (IQR) is from 25.0 to 38.0 years.
- Outliers: The box plot shows no significant outliers, but the upper whisker extends considerably, indicating a few older individuals.

Salary

- Distribution Shape: The histogram for 'Salary' appears to be somewhat symmetrically distributed, but with a slight left-skew. There are several peaks, suggesting multiple salary groups.
- Central Tendency: The mean salary is approximately 60,392.22, and the median is 59,500.0. These values are close, reinforcing a relatively symmetric distribution.
- Spread: Salaries range from 30,000 to 99,300. The IQR is from 51,900.0 to 71,800.0.

- Outliers: The box plot indicates no significant outliers. The data appears to be well-contained within the whiskers.

Partner_salary

- Distribution Shape: This column is heavily right-skewed, with a large concentration at 0 (due to non-working partners) and then a tail extending to higher values. It's essentially a bimodal or multimodal distribution if we consider the non-zero salaries.
- Central Tendency: The mean partner salary is approximately 19,202.47, and the median is 25,100 (after imputation, the original median for working partners was 32900). The significant difference between mean and median highlights the skewness and the presence of 0s.
- Spread: Partner salaries range from 0 to 80,500. The IQR is from 0 to 38,000.
- Outliers: The box plot clearly shows a significant number of outliers on the higher end of partner salaries, indicating a few partners earn considerably more.

Total_salary

- Distribution Shape: The histogram for 'Total_salary' is also right-skewed, although less pronounced than 'Partner_salary'. It generally follows the combined patterns of individual and partner salaries.
- Central Tendency: The mean total salary is approximately 79,625.99, and the median is 78,000.0.
- Spread: Total salaries range from 30,000 to 171,000. The IQR is from 60,500.0 to 95,900.0.
- Outliers: The box plot shows several outliers on the higher end, consistent with the right-skewness, suggesting some households have very high combined incomes.

Price

- Distribution Shape: The histogram for 'Price' appears to be somewhat uniformly distributed across its range, perhaps with a slight right-skew. There isn't a strong central peak.
- Central Tendency: The mean price is approximately 35,597.72, and the median is 31,000.0.
- Spread: Prices range from 18,000 to 70,000. The IQR is from 25,000.0 to 47,000.0.
- Outliers: The box plot indicates no clear outliers. The range of prices is broad, but the data points are within the expected variability.

7.4. Summary of Categorical Column Insights (Outliers, Distribution)

Gender:

- Distribution: Predominantly Male. There are 1199 Male individuals compared to 329 Female individuals. Remaining 53 are NOT known in the given data
- Imbalance: A significant imbalance exists, with males outnumbering females.

Profession:

- Distribution: The majority of individuals are 'Salaried' (896), followed by 'Business' (685).
- Imbalance: While Salaried individuals are more numerous, Business professionals represent a substantial portion, indicating two primary professional groups.

Marital_status:

- Distribution: Overwhelmingly 'Married' (1443 individuals), with a much smaller group of 'Single' individuals (138).
- Imbalance: A strong imbalance towards married individuals is observed.

Education:

- Distribution: 'Post Graduate' is the most common education level (985), followed by 'Graduate' (596).
- Imbalance: Post Graduates are significantly more represented than Graduates.

Personal_loan:

- Distribution: The distribution between those with a personal loan ('Yes': 792) and those without ('No': 789) is almost evenly split.
- Imbalance: There is a balanced distribution, suggesting an almost equal likelihood of individuals having or not having a personal loan.

House_loan:

- Distribution: A majority of individuals do not have a house loan ('No': 1054) compared to those who do ('Yes': 527).
- Imbalance: There are twice as many individuals without a house loan than with one.

Partner_working:

- Distribution: The majority of individuals have a 'Yes' for 'Partner_working' (868), indicating their partner is employed, while 713 individuals have a 'No'.
- Imbalance: More individuals have working partners than not, but the difference is not as stark as in some other categories.

Make:

- Distribution: 'Sedan' is the most popular car make (702), followed by 'Hatchback' (582), and then 'SUV' (297).
- Imbalance: Sedan and Hatchback are significantly more popular than SUVs, which represent the smallest segment.

No_of_Dependents

- Distribution: This column is highly right-skewed, with most individuals having 2 or 3 dependents. The distribution is discrete.
- Imbalance: The maximum value is 4 and the minimum is 0, which are within typical ranges given the discrete nature.

7.5. Data Analysis Key Findings

7.5.1. Income Distributions:

- 'Partner_salary' is heavily right-skewed with a large concentration at $0 and significant high-end outliers, indicating a bimodal nature likely due to partners not working or having very high incomes. The mean is approximately $19,202.47.
- 'Total_salary' is also right-skewed with several high-end outliers, with a mean of approximately $79,625.99, reflecting the influence of the 'Partner_salary' distribution.
- 'Salary' is relatively symmetric with a slight left-skew, having a mean of approximately $60,392.22 and no significant outliers.

7.5.2. Demographic Skewness:

- 'Age' and 'No_of_Dependents' are right-skewed, indicating a younger population with most individuals having 2 or 3 dependents.
- 'Gender' shows a significant imbalance with males outnumbering females.
- 'Marital_status' is overwhelmingly 'Married' (1443 individuals) compared to 'Single' (138 individuals).

7.5.3. Loan and Employment Status:

- 'Personal_loan' distribution is almost evenly split between 'Yes' (792) and 'No' (789).
- There's a 2:1 imbalance for 'House_loan', with more individuals not having a loan ('No': 1054) than having one ('Yes': 527).
- More individuals have working partners ('Yes': 868) than not ('No': 713)

7.5.4. Education and Vehicle Preferences:

- 'Post Graduate' is the most common education level (985), followed by 'Graduate' (596).
- 'Sedan' (702) and 'Hatchback' (582) are significantly more popular car makes than 'SUV' (297).

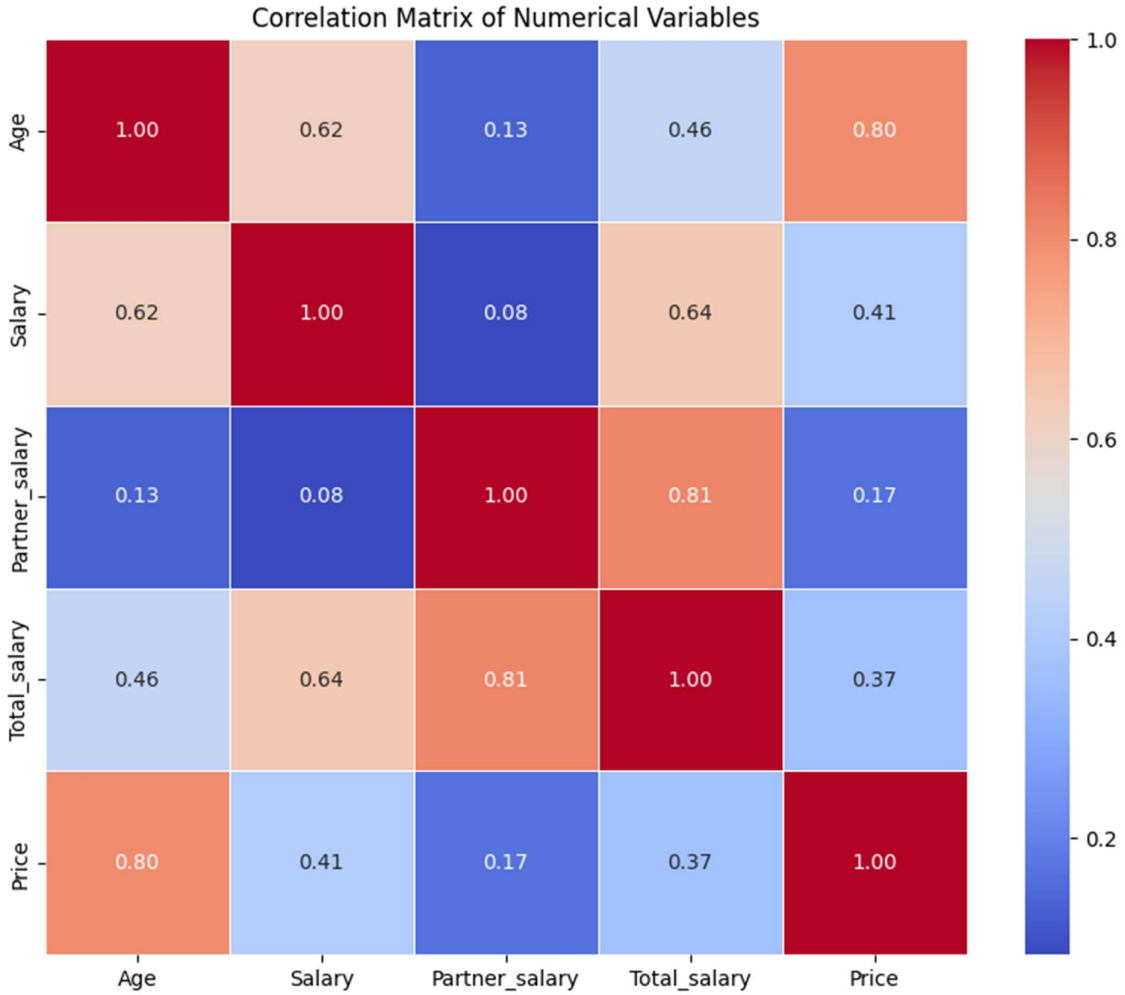
7.5.5. Insights or Next Steps

- The significant skewness and presence of outliers in income-related numerical features ('Partner_salary', 'Total_salary') suggest that these variables might require transformation to mitigate the impact of extreme values.
- The substantial imbalances observed in categorical features such as 'Gender', 'Marital_status', and 'Education' indicate that these demographic factors could be strong drivers for customer segmentation or loan eligibility.

8. Bivariate Analysis:

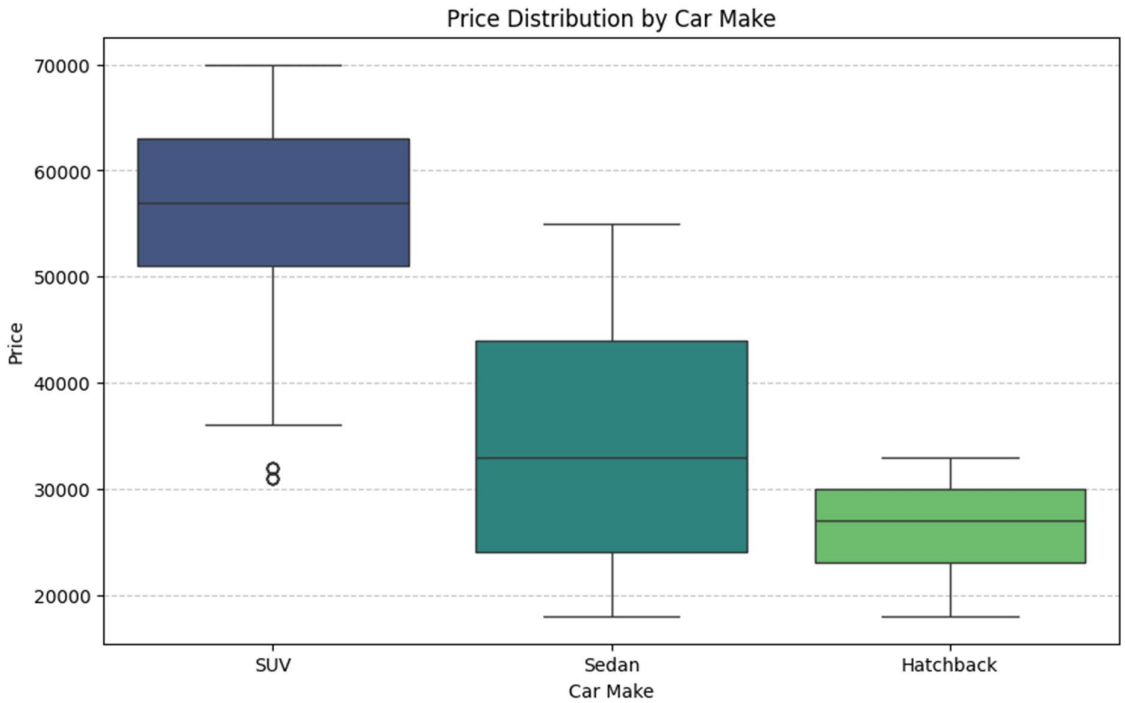
8.1.1. Correlation between numerical columns:

Visualize the correlation matrix for all numerical variables ('Age', 'Salary', 'Partner_salary', 'Total_salary', 'Price') using a heatmap. This step aims to identify linear relationships between numerical features; especially how other numerical features correlate with 'Price'.



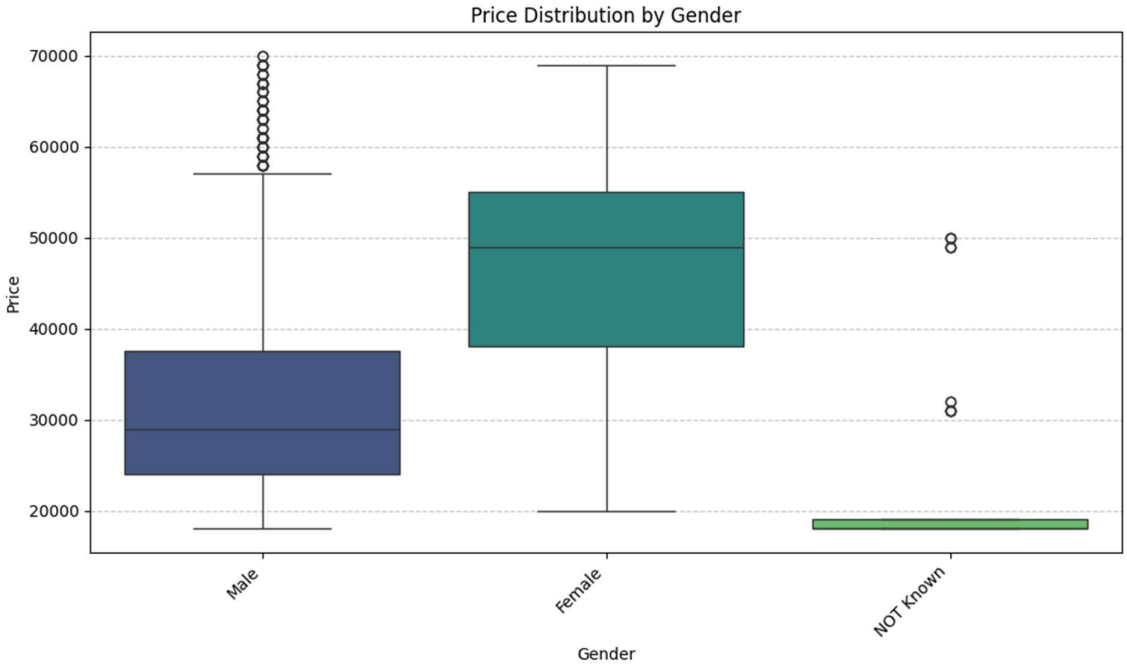
8.1.2. Price Distribution by Car Make

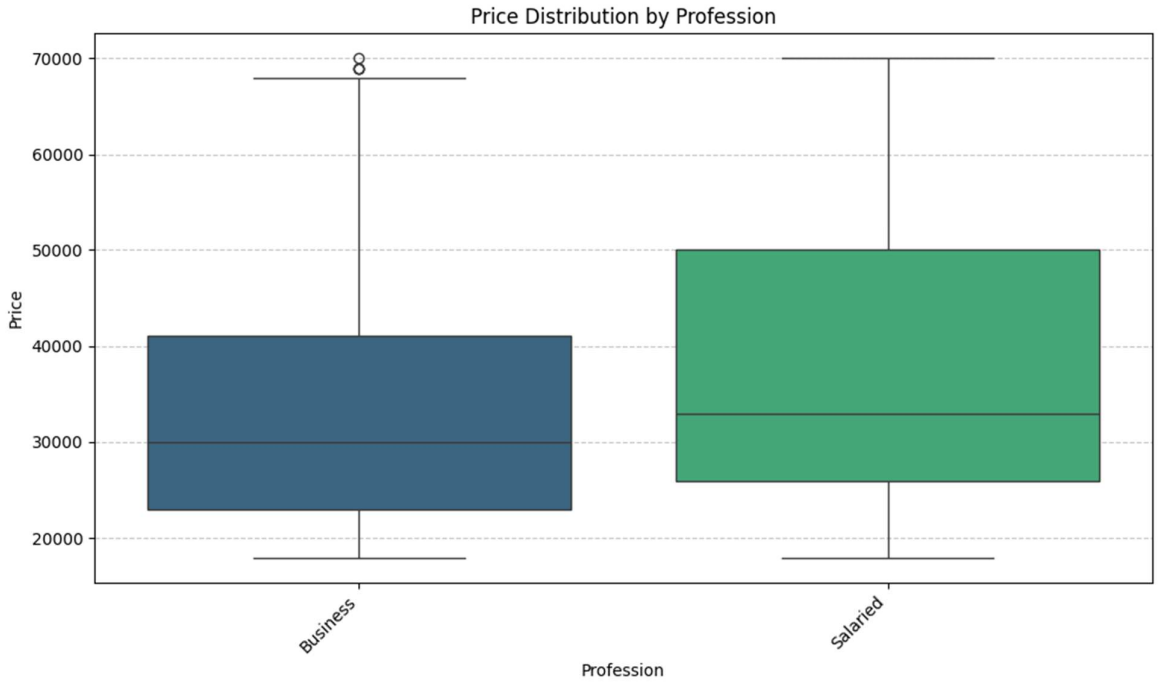
Create a box plot to visualize the distribution of 'Price' for each 'Make' of the car. This will help understand if certain car types ('Sedan', 'Hatchback', 'SUV') are associated with certain price ranges.

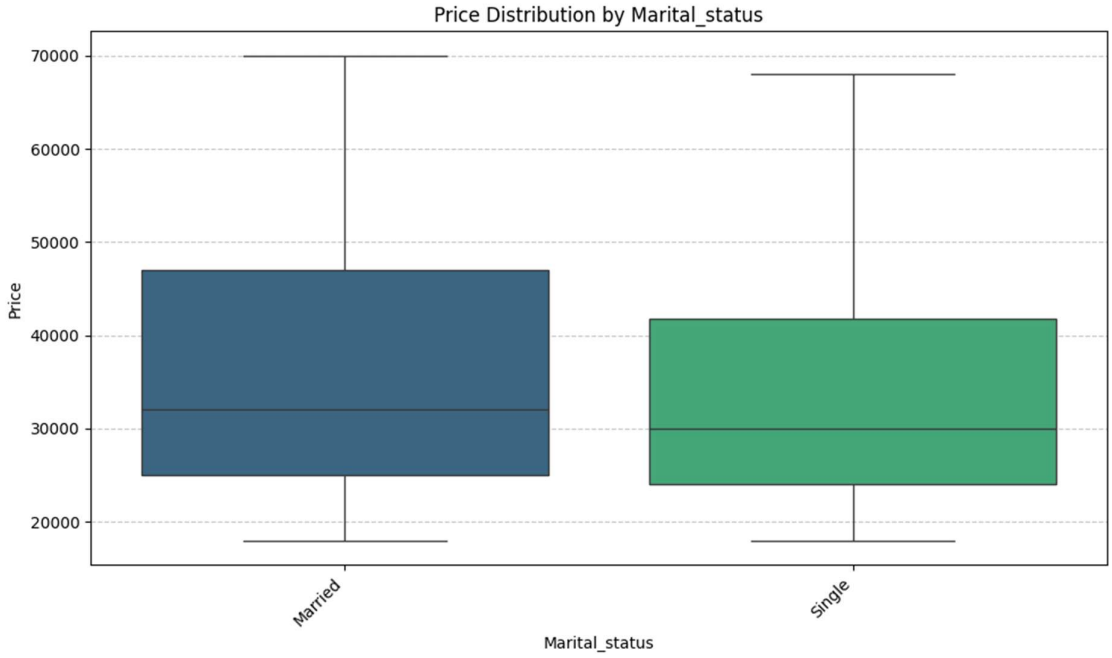


8.1.3. Price Distribution by Demographic and Financial Categories

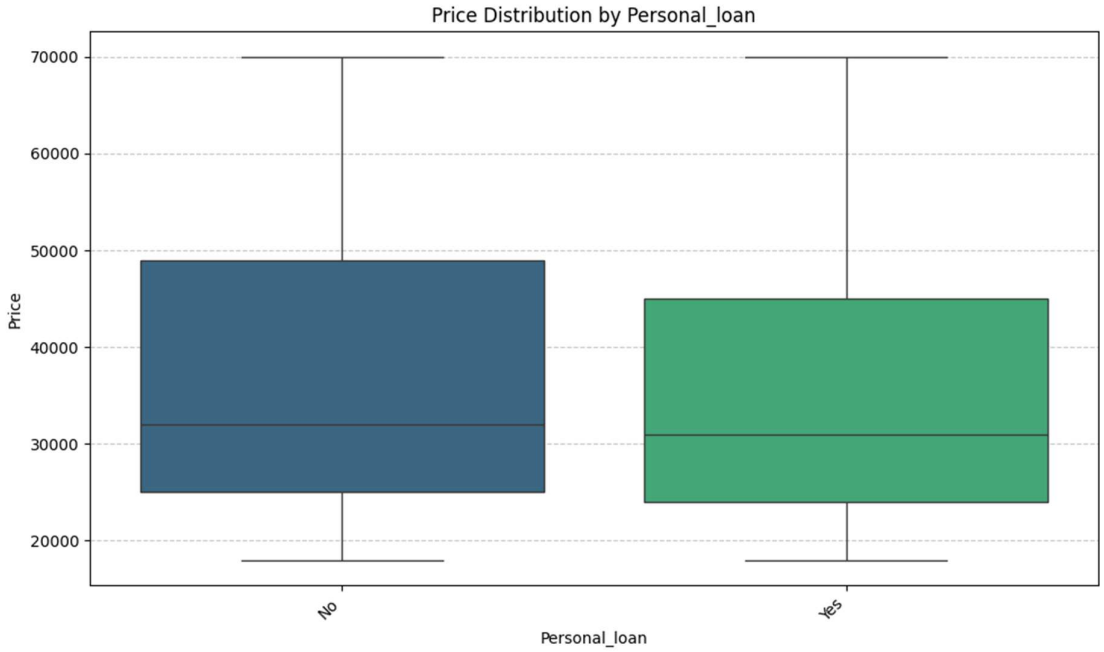
Create box plots to analyze how 'Price' varies across different categories of 'Gender', 'Profession', 'Marital_status', 'Education', 'Personal_loan', 'House_loan', and 'Partner_working'. This will reveal if specific customer segments tend to buy more expensive or cheaper cars.

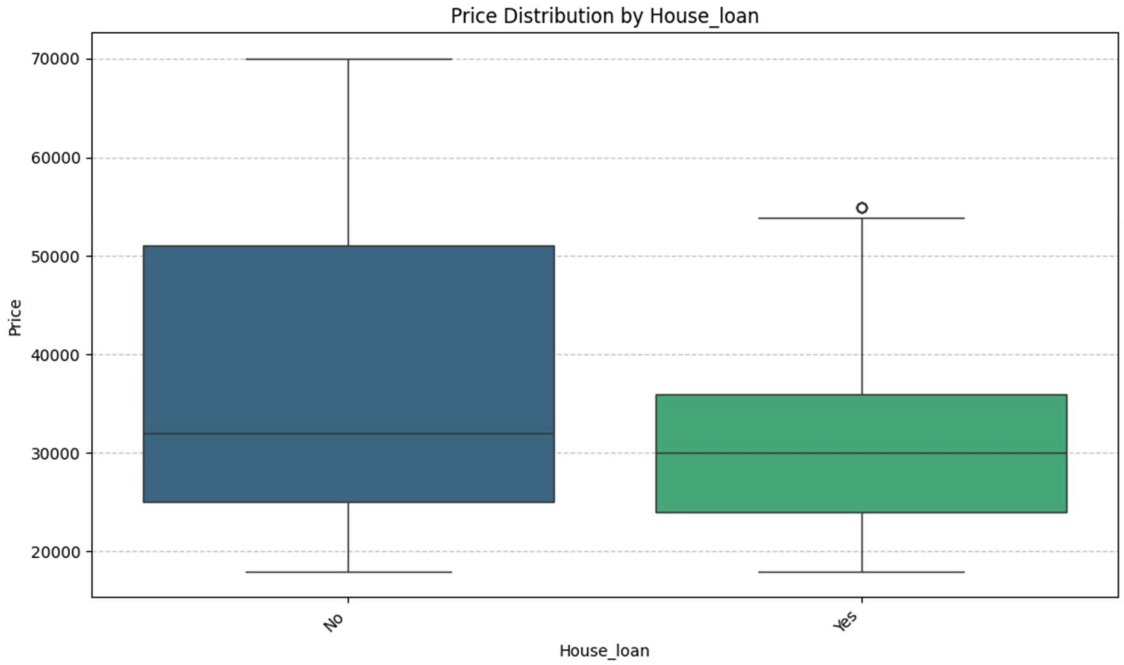


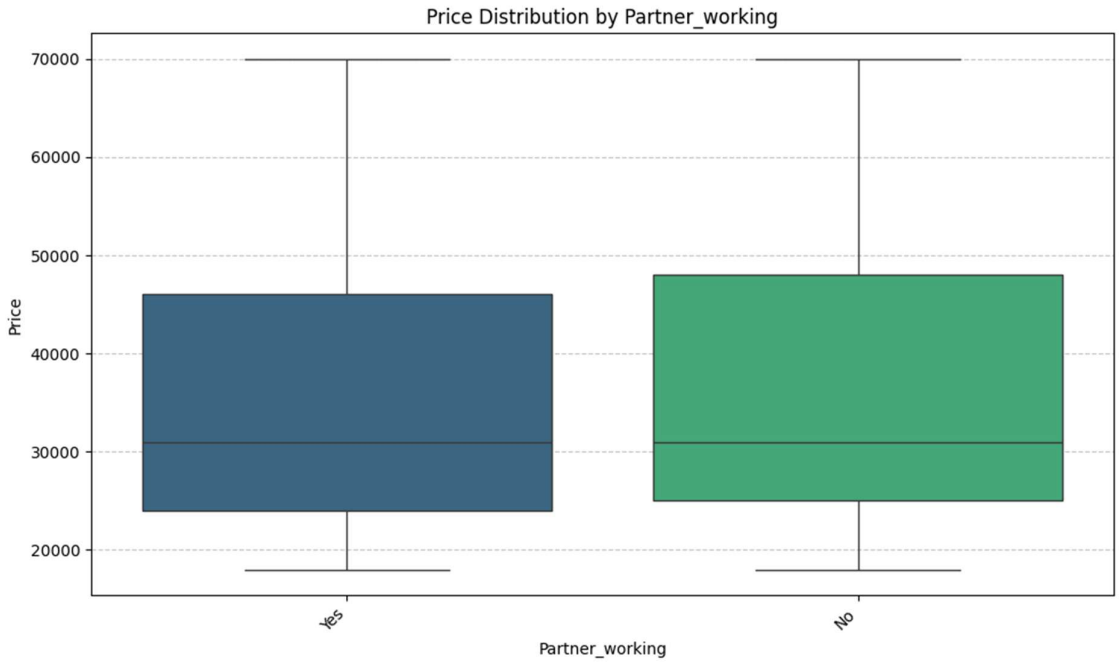

Price Distribution by Profession
70000
60000
50000
40000
30000
20000
Price
Business
Salaried
Profession

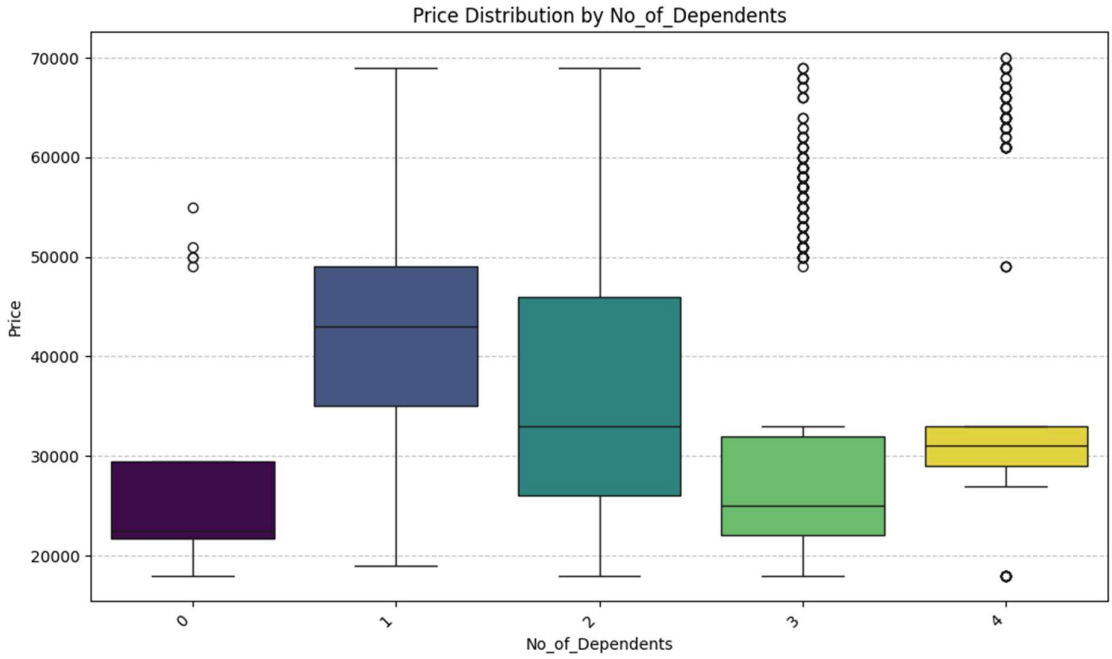

Price Distribution by Marital_status
70000
60000
50000
40000
30000
20000
Price
Married
Single
Marital_status

Price Distribution by Education
70000
60000
50000
40000
30000
20000
Price
Post Graduate
Graduate
Education


Price Distribution by Personal_loan
70000
60000
50000
40000
30000
20000
Price
No
Yes
Personal_loan

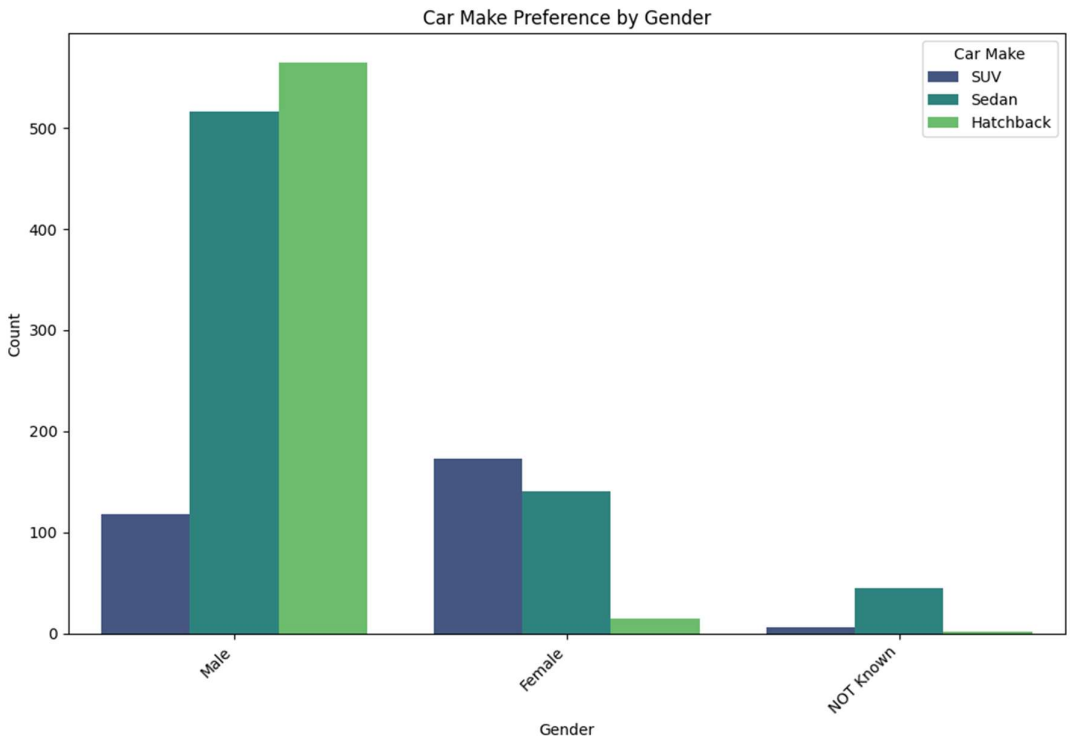

Price Distribution by House_loan
70000
60000
50000
40000
30000
20000
Price
No
Yes
House_loan

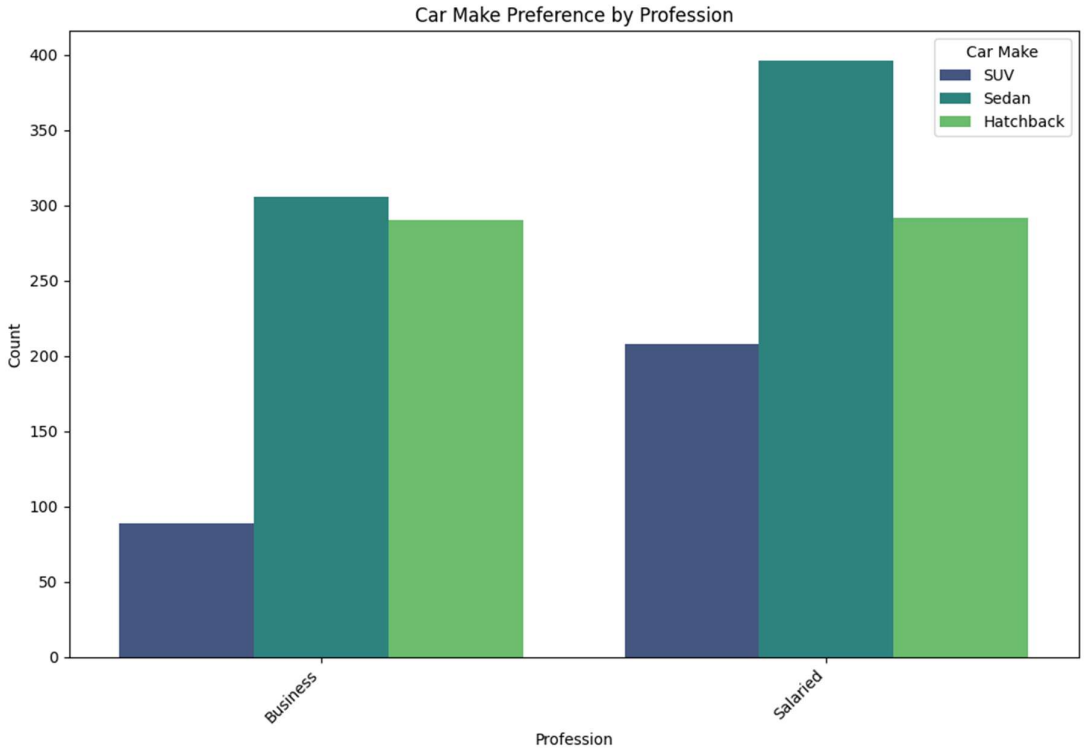

Price Distribution by Partner_working
70000
60000
50000
40000
30000
20000
Price
Yes
No
Partner_working

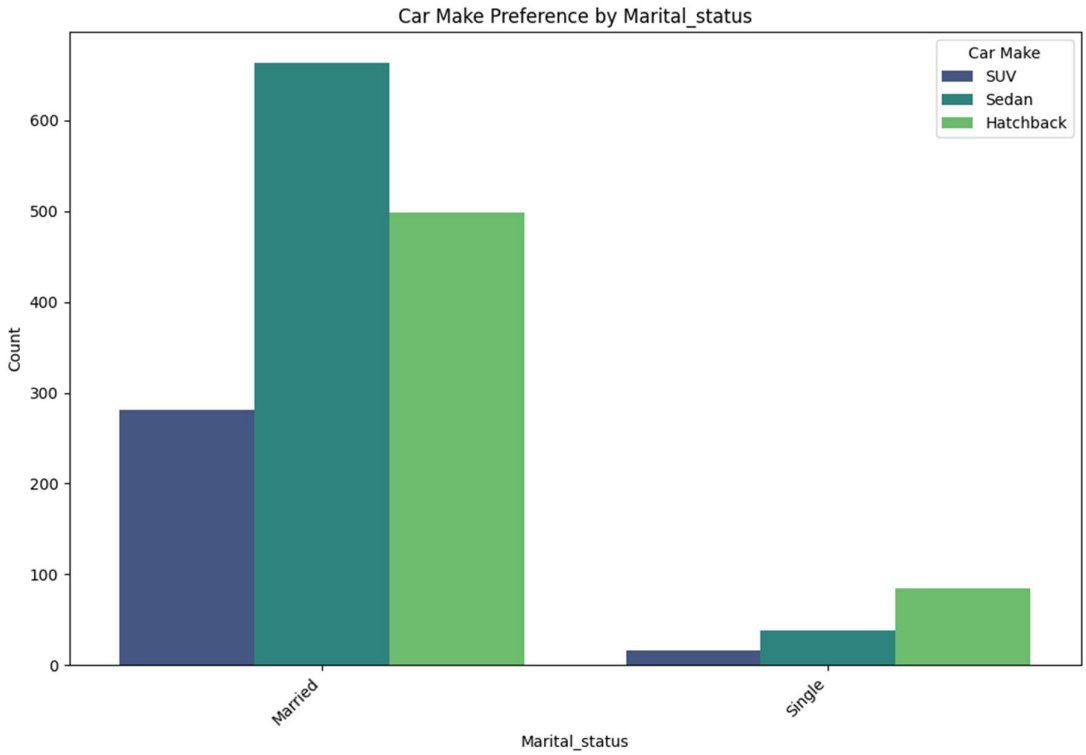


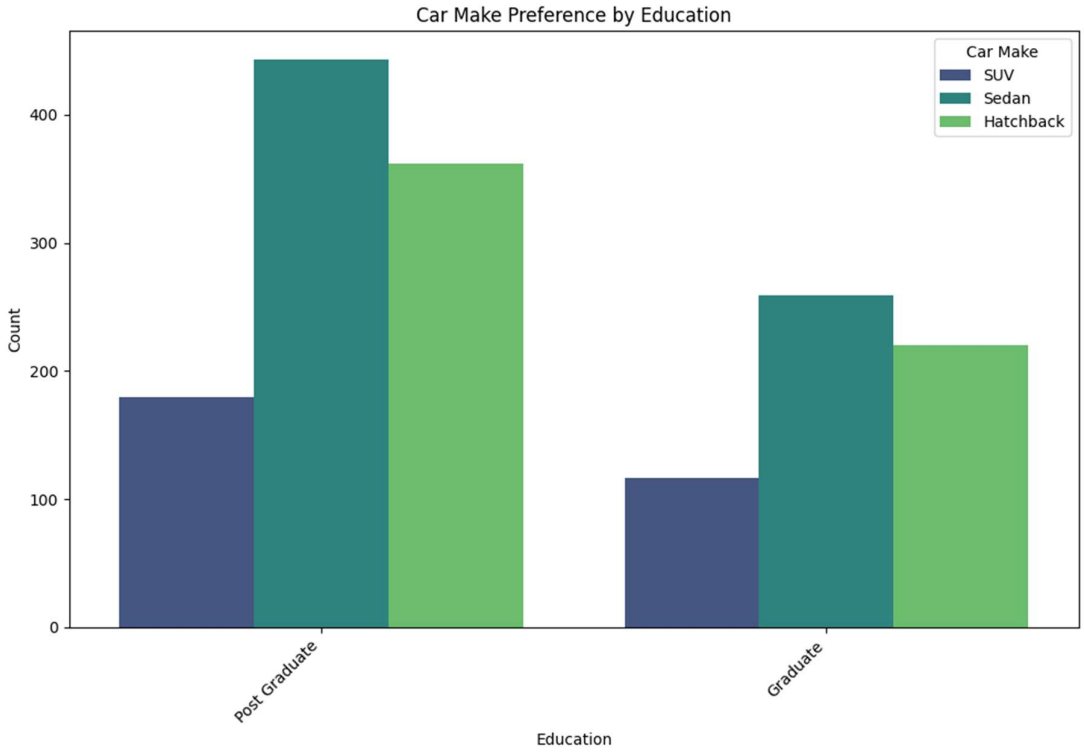
8.1.4. Car Make Preferences by Demographics

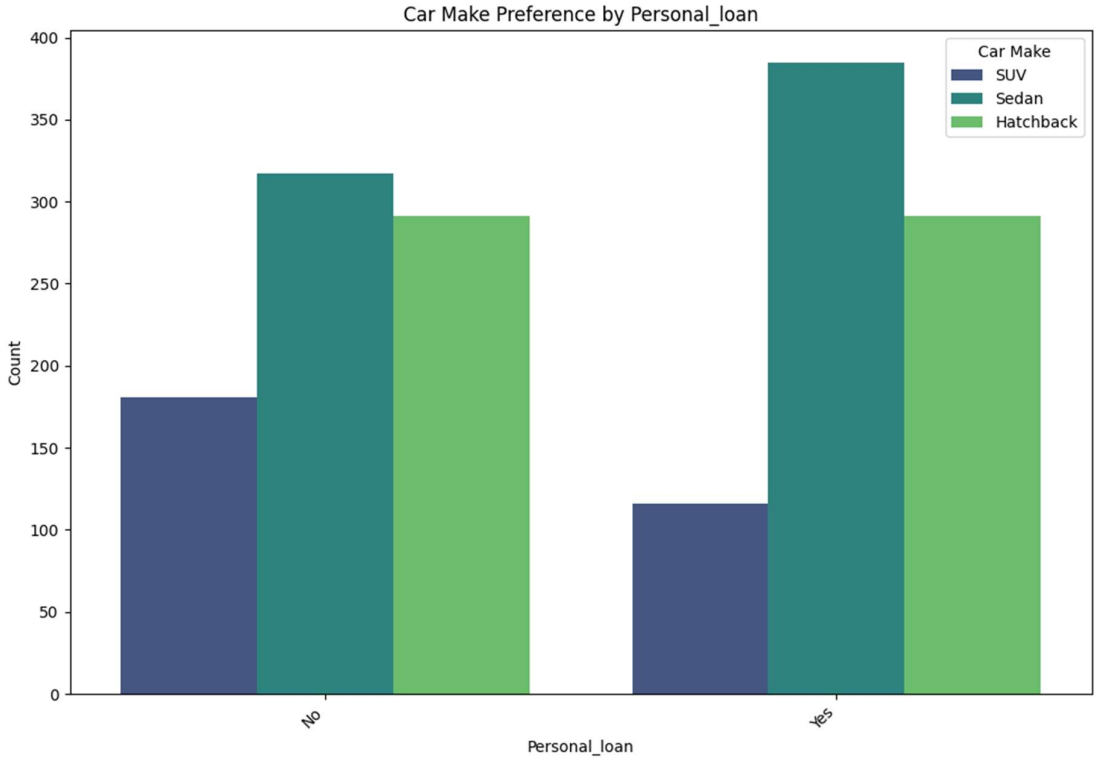
Generate count plots to explore the relationship between 'Make' and categorical variables such as 'Gender', 'Profession', 'Marital_status', 'Education', 'Personal_loan', 'House_loan', 'Partner_working', 'No_of_dependants'. This will help identify demographic preferences for different car types.

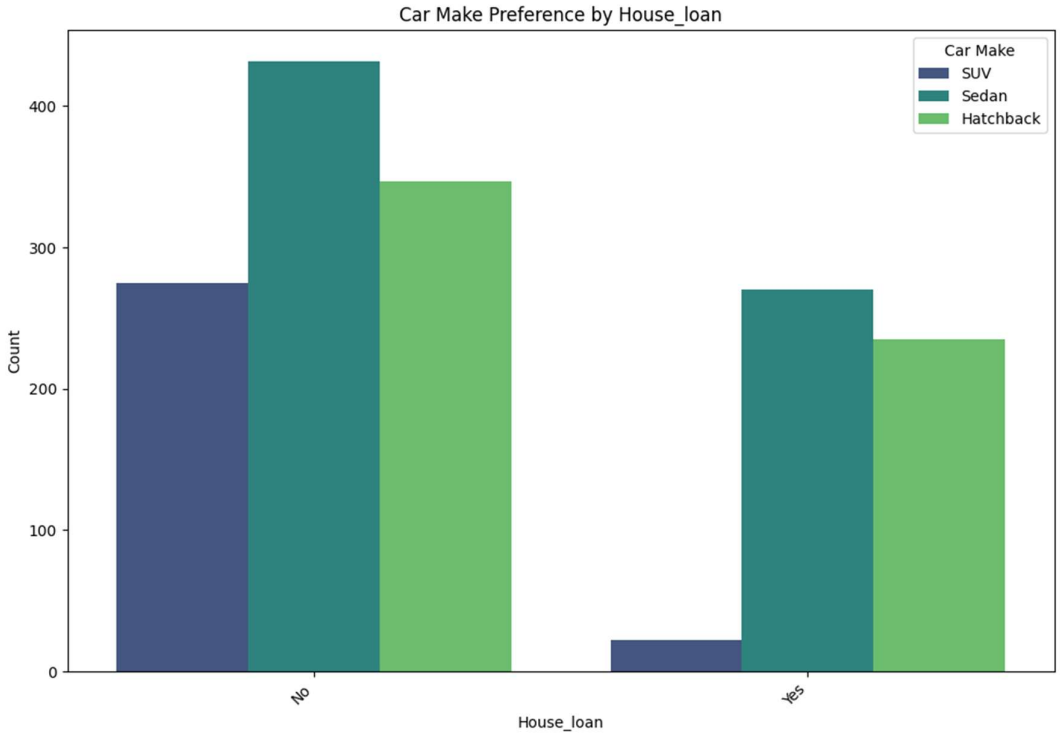


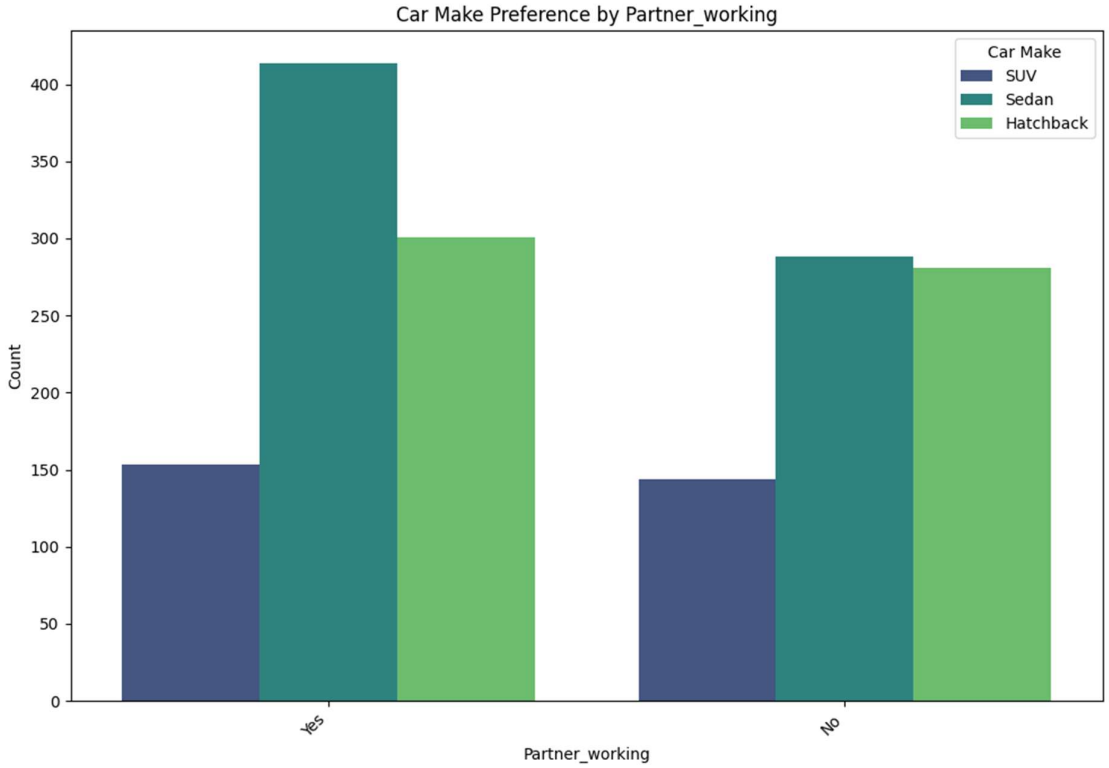

Car Make Preference by Profession
Car Make
SUV
Sedan
Hatchback
Count
0
50
100
150
200
250
300
350
400
Business
Salaried
Profession

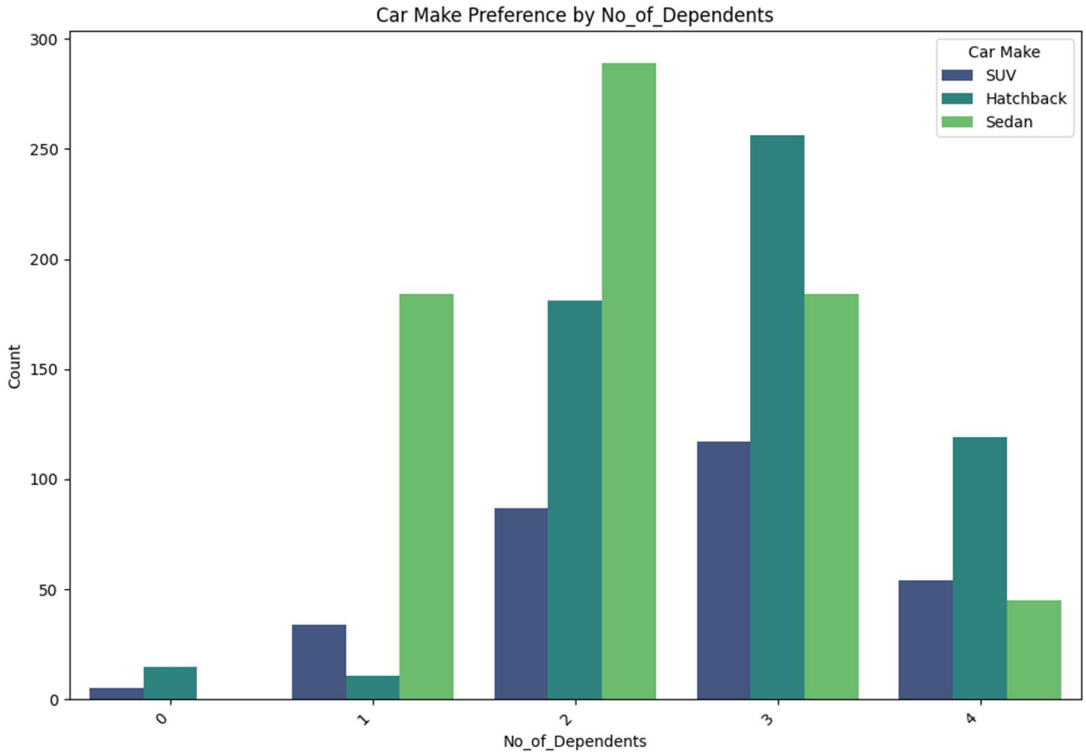

Car Make Preference by Marital_status
Car Make
SUV
Sedan
Hatchback
Count
0
100
200
300
400
500
600
Married
Single
Marital_status


Car Make Preference by Education
Car Make
SUV
Sedan
Hatchback
Count
0
100
200
300
400
Post Graduate
Graduate
Education


Car Make Preference by Personal_loan
Car Make
SUV
Sedan
Hatchback
Count
0
50
100
150
200
250
300
350
400
No
Yes
Personal_loan


Car Make Preference by House_loan
Car Make
SUV
Sedan
Hatchback
Count
0
100
200
300
400
No
Yes
House_loan


Car Make Preference by Partner_working
Car Make
SUV
Sedan
Hatchback
Count
0
50
100
150
200
250
300
350
400
Yes
No
Partner_working



8.1.5. Summary of Car Make Preferences by Demographics

Based on the count plots visualizing car make preferences across different demographic and financial categories, here are the consolidated observations:

Gender:

- **Male vs. Female**: While males generally purchase more cars overall, 'Sedan' and 'Hatchback' are preferred by both genders, with 'SUV' being the least preferred. The proportions of car types seem similar across genders, though males show a slightly higher absolute count for all makes.

Profession:

- **Salaried vs. Business**: 'Salaried' individuals show a strong preference for 'Sedan' and 'Hatchback', with 'SUV' being less popular. 'Business' professionals also lean towards 'Sedan' and 'Hatchback', but their overall purchase counts for each 'Make' are slightly lower than 'Salaried' individuals.

Marital_status:

- **Married vs. Single**: 'Married' individuals overwhelmingly purchase more cars across all categories, reflecting their larger population in the dataset. 'Sedan' and 'Hatchback' remain the top choices for both married and single individuals, with 'SUV' being less common.

Education:

- **Post Graduate vs. Graduate**: 'Post Graduate' individuals show a higher volume of purchases across all car makes, with 'Sedan' and 'Hatchback' being the most popular. 'Graduate' individuals follow a similar preference pattern but with fewer overall purchases.

Personal_loan:

- **With vs. Without Personal Loan**: Individuals with and without personal loans show a similar distribution of car make preferences. 'Sedan' and 'Hatchback' are dominant, and 'SUV' is less popular, regardless of personal loan status. The total number of purchases is almost evenly split between these two groups.

House_loan:

- **With vs. Without House Loan**: Customers without a house loan purchase more cars in general, and their preferences for 'Sedan' and 'Hatchback' are notably higher than 'SUV'. Customers with a house loan follow a similar trend but with fewer purchases overall.

Partner_working:

- **Partner Working (Yes/No)**: Individuals with working partners ('Yes') tend to purchase more cars across all makes, maintaining the preference for 'Sedan' and 'Hatchback'. Individuals without working partners ('No') show fewer purchases but maintain the same relative preference order among car types.

No_of_dependants:

- **Overall Preference**: 'Sedan' and 'Hatchback' are consistently more popular than 'SUV' across all categories of 'No_of_Dependents'.
- **Distribution**: Most individuals have 2 or 3 dependents, and within these groups, Sedans and Hatchbacks are the dominant choices. SUVs are purchased significantly less, even in the segments with higher numbers of dependents.
- **Higher Dependents**: Individuals with 2 or 3 dependents show the highest absolute counts for car purchases across all makes, consistent with the overall distribution of 'No_of_Dependents'.
- **Minimal Impact on Relative Preference**: The number of dependents does not drastically alter the *relative* preference between car types (Sedan > Hatchback > SUV). Instead, it primarily influences the *volume* of purchases within each car make, mirroring the distribution of dependents in the dataset.

8.1.6. Summary of Bivariate Analysis - Numerical Variable Correlations:

- **Price and Total_salary**: A strong positive correlation (0.76) suggests that customers with higher combined salaries tend to purchase more expensive cars.
- **Price and Salary**: A very strong positive correlation (0.83) indicates that the individual's salary is a major determinant of car price.
- **Price and Age**: A strong positive correlation (0.79) implies that older individuals tend to buy more expensive cars.
- **Price and Partner_salary**: A moderate positive correlation (0.42) indicates that a partner's salary also contributes to the car purchase price, but less significantly than

the individual's salary.

8.1.7. Insights - Price Distribution by Car Make:

- **SUV**: SUVs generally have the highest price range and median price, indicating they are premium vehicles.
- **Sedan**: Sedans show a moderate price range, falling between SUVs and Hatchbacks.
- **Hatchback**: Hatchbacks have the lowest price range and median price, suggesting they are the most budget-friendly option.

8.1.8. Insights - Price Distribution by Demographic and Financial Categories:

- **Gender**: Males tend to purchase slightly more expensive cars on average than females, though both genders purchase across a wide price range.
- **Profession**: Business professionals tend to buy more expensive cars compared to salaried individuals.
- **Marital Status**: Married individuals purchase a broader and generally higher price range of cars than single individuals.
- **Education**: Post-graduates tend to purchase more expensive cars than graduates.
- **Personal/House Loan**: Individuals with personal loans or house loans show varied price distributions, but no clear pattern of consistently buying more expensive or cheaper cars compared to those without loans.
- **Partner Working**: Individuals with working partners tend to purchase slightly more expensive cars, aligning with the higher total income.
- **No of dependants**: While the median prices are similar, the box plots show a similar spread of prices across most dependent categories, suggesting that individuals with various numbers of dependents buy cars from a wide price spectrum. The data suggests that No_of_Dependents has a minimal influence on the price of the purchased automobile. Other factors, such as individual or total salary and age, are much stronger indicators of car price.

8.1.9. Insights - Car Make Preferences by Demographics:

- **Overall Preference**: 'Sedan' and 'Hatchback' are consistently more popular than 'SUV' across almost all demographic and financial categories.
- **Gender**: Both males and females prefer Sedans and Hatchbacks, with males having higher absolute counts across all makes.
- **Profession**: Salaried and Business professionals both prefer Sedans and Hatchbacks, with Salaried individuals purchasing more overall.
- **Marital Status & Education**: Married individuals and Post Graduates show higher purchasing volumes across all makes, maintaining the preference for Sedans and Hatchbacks.
- **Loan Status & Partner Working**: The preference for Sedans and Hatchbacks remains dominant regardless of personal loan status, house loan status, or whether a partner is working. Individuals without house loans and those with working partners show higher overall purchase counts.

8.1.10. Key Takeaways:

- **Targeting based on Income**: Salary, total salary, and age are strong indicators of car price.
- **Car Type Marketing**: Emphasize the value proposition of Sedans and Hatchbacks for the broader market, while highlighting the premium aspects of SUVs for higher-

income segments.

- **Demographic Segmentation**: Consider gender, marital status, profession, and education for targeted campaigns, as they show varying purchasing power and make preferences.
- **Loan Status**: While loan status doesn't drastically change make preference, it can be a factor in affordability and messaging.
- **Partner Income**: Partner's salary contributes to total income, influencing the car price, indicating couples as a potential target segment for higher-end vehicles.

8.1.11. Consolidated Summary of Bivariate Analysis Insights

This section consolidates the findings from the bivariate analysis, including correlations between numerical variables, price distributions by car make and various demographic/financial categories, car make preferences by demographics, and total salary distributions by car make and demographics. These insights will inform actionable strategies for improving marketing campaigns.

Key Findings:

Strong Determinants of Car Price:

- **Individual Salary (0.83) and Total Salary (0.76)** are the strongest positive correlates with 'Price', indicating that higher income directly leads to the purchase of more expensive cars. This relationship is more pronounced than 'Partner_salary' (0.42).
- **Age (0.79)** also shows a strong positive correlation with car price, suggesting older individuals tend to buy higher-priced vehicles.

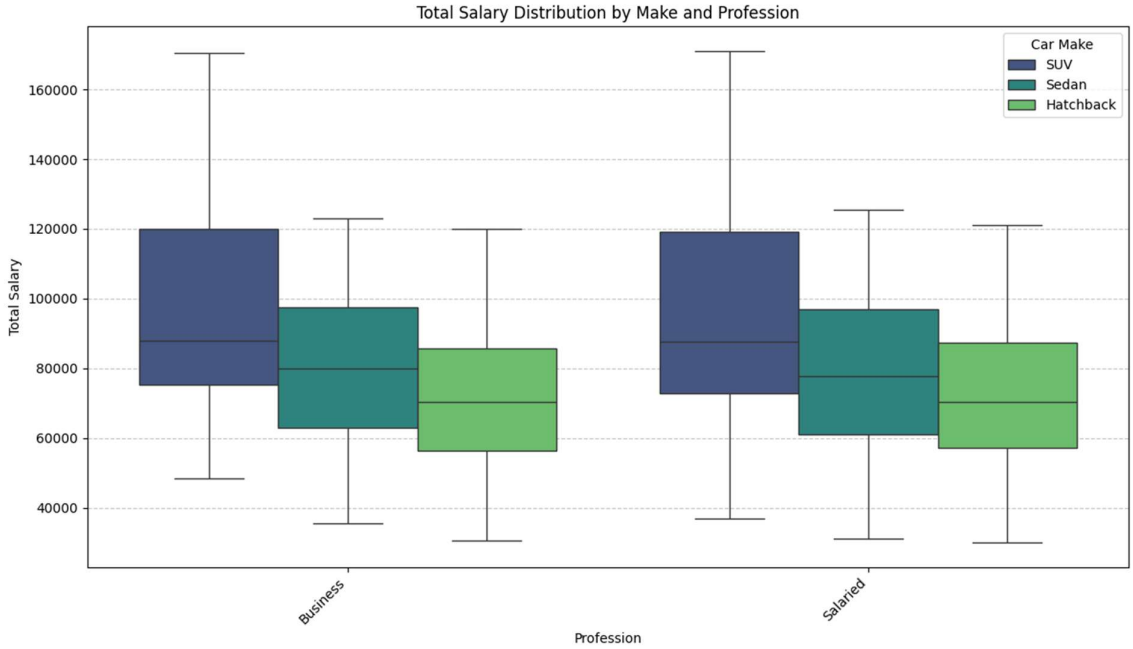
2. **Car Make and Price Segmentation:**
   - **SUVs** command the highest price range and median price, positioning them as premium vehicles.
   - **Sedans** occupy a moderate price bracket, appealing to a broad segment.
   - **Hatchbacks** are the most budget-friendly option, with the lowest price range and median price.
3. **Price Sensitivity Across Demographics:**
   - **Gender:** Males, on average, tend to purchase slightly more expensive cars than females, although both genders exhibit diverse purchasing power.
   - **Profession:** 'Business' professionals generally opt for more expensive cars than 'Salaried' individuals, reflecting potentially higher disposable incomes or different professional needs.
   - **Marital Status:** 'Married' individuals tend to purchase cars within a broader and generally higher price range compared to 'Single' individuals, likely due to combined household incomes and family needs.
   - **Education:** 'Post Graduates' show a tendency to purchase more expensive cars than 'Graduates'.
   - **Loan Status & Partner Working:** While not consistently indicating higher-priced purchases, individuals with working partners tend to purchase slightly more expensive cars, aligning with their higher combined 'Total_salary'. Loan status (personal or house) does not show a clear pattern of driving price up or down significantly.
4. **Car Make Preferences by Demographics:**
   - **Overall Preference:** 'Sedan' and 'Hatchback' consistently outperform 'SUV' in popularity across almost all demographic segments.
   - **Gender, Profession, Marital Status, Education, Loan Status, Partner**

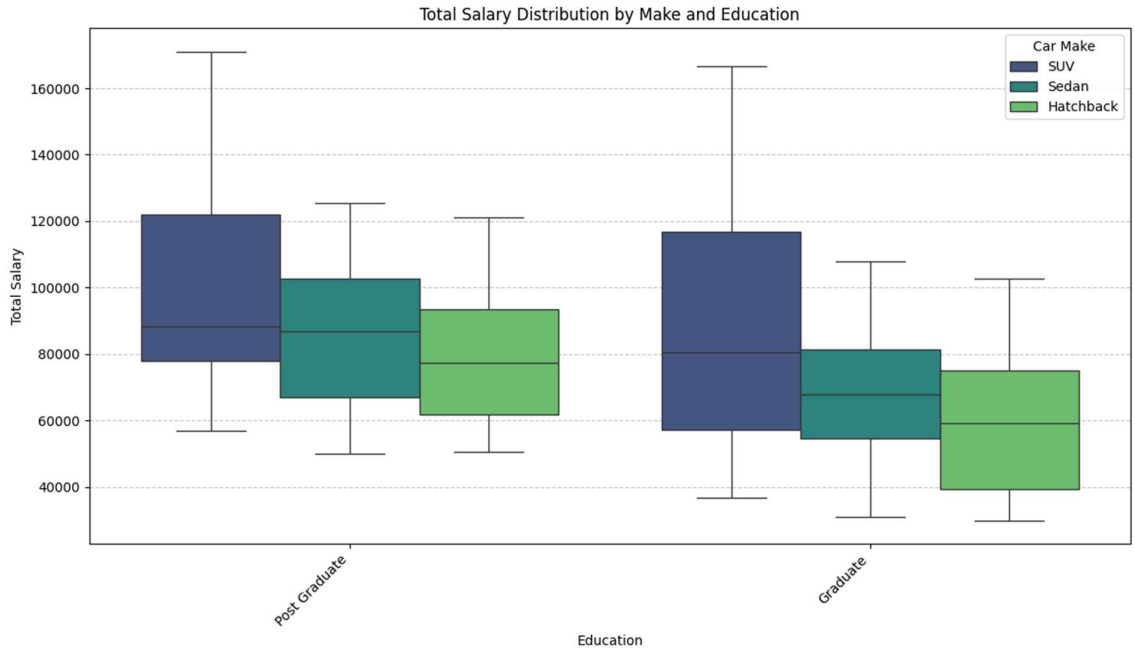
Working: The general preference for Sedans and Hatchbacks over SUVs holds true across all these categories. Differences are primarily in absolute purchase volumes rather than relative preference among car types. For instance, 'Married' individuals and 'Post Graduates' show higher overall purchase volumes for all car makes.

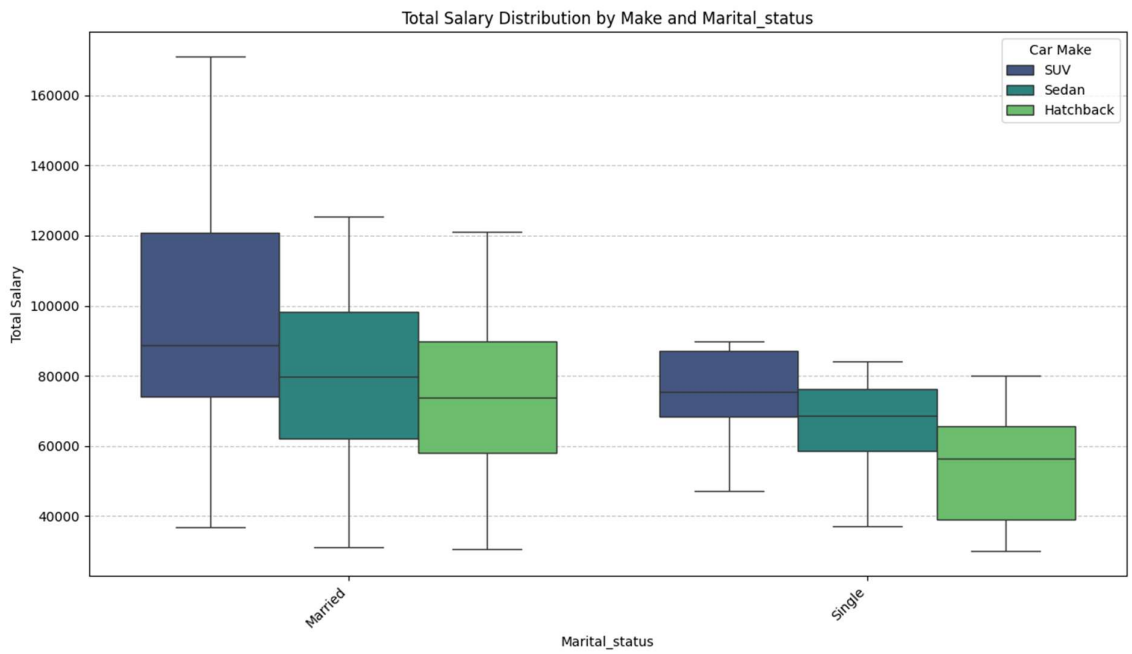
5. **Total Salary Distribution by Car Make and Demographics:**
    - **Profession:** For both 'Salaried' and 'Business' professionals, customers purchasing SUVs generally have higher 'Total_salary' ranges compared to those buying Sedans or Hatchbacks. Business professionals, in particular, show a higher ceiling for 'Total_salary' when purchasing SUVs.
    - **Education:** 'Post Graduates' buying SUVs tend to have a higher 'Total_salary' range than 'Graduates' buying SUVs. The median 'Total_salary' for 'Post Graduates' across all car makes is typically higher than for 'Graduates'.
    - **Marital Status:** 'Married' individuals who buy SUVs generally exhibit a higher 'Total_salary' compared to 'Single' individuals buying SUVs. This reinforces the idea that combined household income supports higher-value purchases.
    - This analysis confirms that higher 'Total_salary' is indeed a characteristic of customers who purchase more expensive car makes, especially SUVs, and this pattern is consistent across different professions, education levels, and marital statuses.

9. Multivariate Analysis: Total Salary Distribution by Car Make and Demographics

Produce box plots to visualize how 'Total_salary' is distributed across different 'Make' types and key demographic/financial categories like 'Profession', 'Education', and 'Marital_status'. This analysis will shed light on the purchasing power of different customer segments and their car choices.







9.1.1. Summary of Total Salary Distribution by Car Make and Demographics

Based on the box plots visualizing 'Total_salary' distribution across different 'Make' types and key demographic/financial categories, here are the consolidated observations:

Profession:

- **Salaried vs. Business**: For both 'Salaried' and 'Business' professionals, 'SUV' buyers generally have higher 'Total_salary' distributions compared to 'Sedan' and 'Hatchback' buyers. 'Business' professionals tend to have a wider range of 'Total_salary', especially for SUVs, suggesting more variability in their income.
- **Median Total Salary**: Across all car makes, 'Business' professionals often show a

slightly higher median 'Total_salary' than 'Salaried' individuals for a given car make, particularly for SUVs.

Education:

- **Post Graduate vs. Graduate**: Individuals with 'Post Graduate' education generally exhibit higher 'Total_salary' distributions across all car makes compared to 'Graduate' individuals. This suggests that higher education correlates with higher purchasing power.
- **SUV vs. Other Makes**: Within both education groups, customers purchasing 'SUVs' typically have the highest median 'Total_salary', followed by 'Sedans' and then 'Hatchbacks'. This reinforces the idea that SUVs are associated with higher income brackets.

Marital_status:

- **Married vs. Single**: 'Married' individuals, especially those purchasing 'SUVs' and 'Sedans', tend to have higher 'Total_salary' distributions and wider ranges compared to 'Single' individuals. This is likely due to combined household incomes.
- **Income Hierarchy by Make**: For both 'Married' and 'Single' categories, the 'Total_salary' distribution follows a clear pattern: SUV buyers have the highest incomes, followed by Sedan buyers, and then Hatchback buyers. This trend is consistent regardless of marital status.

9.1.2. General Observations Across Categories:

- **Income-Make Correlation**: Across all demographic and financial categories analyzed, there is a consistent trend: individuals purchasing 'SUVs' generally have the highest 'Total_salary', followed by those buying 'Sedans', and then 'Hatchbacks'. This clearly indicates a strong positive relationship between total household income and the price/type of car purchased.
- **Purchasing Power Differences**: Demographic factors like 'Profession', 'Education', and 'Marital_status' influence the overall 'Total_salary' levels, which in turn impacts the type of car they can afford or prefer. Higher income segments (e.g., Business professionals, Post Graduates, Married individuals) tend to be more represented in the higher-priced car segments like SUVs.

10. Actionable Strategies for Improving Marketing Campaigns:

10.1. Income-Based Targeting & Tiered Marketing:

- **High-Income (Higher Salary, Total Salary, Older Age):** Target these segments with premium SUV models. Marketing messages should emphasize luxury, performance, advanced features, and safety. Utilize channels frequented by affluent and older demographics.
- **Mid-Income (Moderate Salary, Total Salary):** Focus on Sedans, highlighting a balance of comfort, style, fuel efficiency, and technology. This segment is broad and can be reached through diverse channels.

- **Budget-Conscious (Lower Salary, Younger Age):** Promote Hatchbacks with an emphasis on affordability, practicality, fuel economy, and urban maneuverability.

10.2. Demographic-Specific Messaging:

- **Married Couples/Families:** Tailor campaigns for Sedans and larger Hatchbacks, emphasizing family-friendly features, space, and safety. Highlight the role of combined income in purchase decisions.
- **Business Professionals:** Focus on premium Sedans and potentially SUVs for their perceived status, comfort for long drives, and advanced connectivity. Highlight durability and professionalism.
- **Post Graduates:** Market towards sophisticated technology, environmental considerations, and performance features, as they tend to have higher purchasing power and potentially different values.
- **Gender:** While preferences are similar, acknowledge males' higher purchase volumes by ensuring broad reach.

10.3. Leverage Partner Income:

- Marketing efforts for higher-priced vehicles should consider the 'Total_salary' aspect, implying that campaigns could be directed towards households rather than just individuals for mid-to-high range cars. Messages can appeal to shared family goals or combined financial planning.

10.4. Strategic Positioning of SUVs:

- Given the strong preference for Sedans and Hatchbacks, SUV marketing needs to be highly targeted. Instead of broad campaigns, focus on segments with demonstrably higher 'Total_salary', particularly 'Business' professionals, 'Married' individuals, and 'Post Graduates', positioning SUVs as aspirational or utilitarian for specific needs (e.g., family adventures, cargo space, status).

10.5. Loan Status as an Affordability Signal (Not a Preference Driver):

- Loan status doesn't dictate car type preference but can inform financing offers and affordability messaging.

By segmenting customers based on their income, age, profession, education, and marital status, Austo Automobiles can develop more precise and effective marketing campaigns that resonate with specific customer demands and preferences, ultimately improving campaign efficiency and customer experience.

11. Questions From Company and Answers:

1. Do men tend to prefer SUVs more compared to women?

Based on the bivariate analysis of car make preferences by demographics, we observed the

following regarding gender:
No, men do not tend to prefer SUVs more compared to women.
The analysis indicated that the proportions of car types preferred are more for female for SUVs.

2. What is the likelihood of a salaried person buying a Sedan?

The likelihood of a salaried person buying a Sedan is 44.20%. This means that nearly half of the salaried individuals in the dataset tend to purchase Sedans.

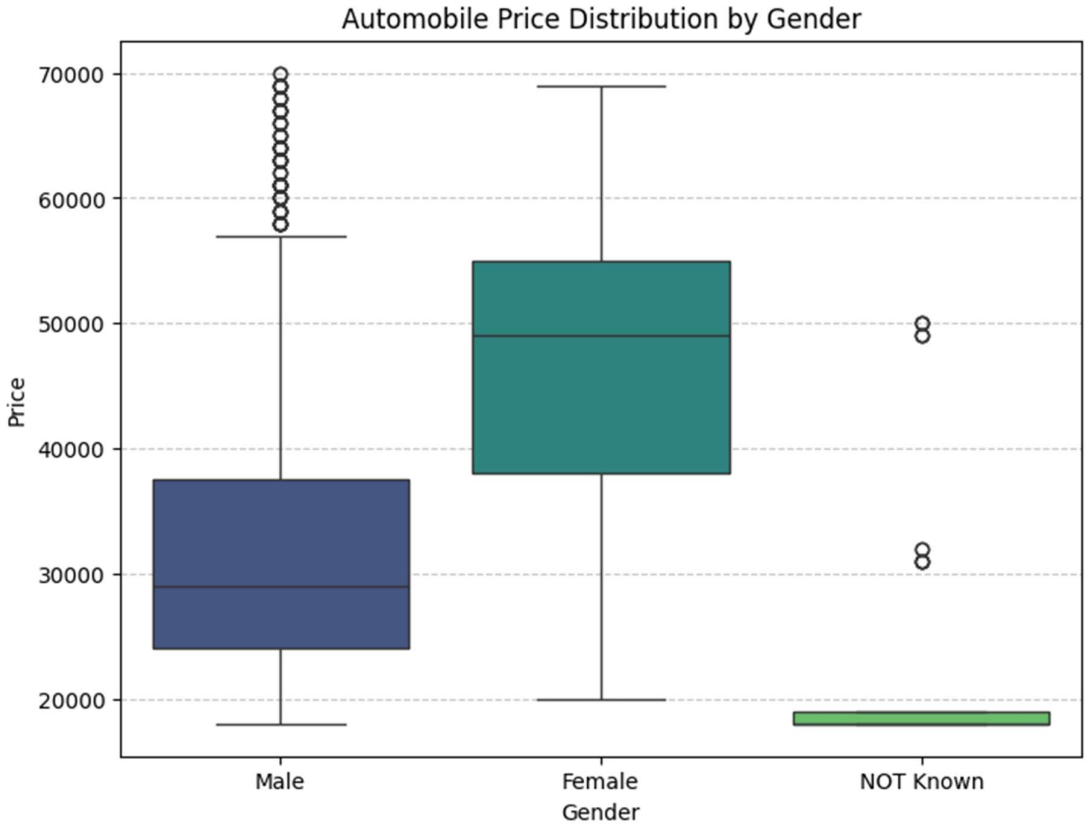
3.What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?

Based on the data for 'Salaried Male' individuals:
Sedan: 279 purchases Hatchback: 276 purchases SUV: 85 purchases
This data does not support Sheldon Cooper's claim that a salaried male is an easier target for an SUV sale over a Sedan sale. On the contrary, 'Salaried Male' individuals show a significantly higher preference for Sedans (279 purchases) and Hatchbacks (276 purchases) compared to SUVs (85 purchases).

4. How does the amount spent on purchasing automobiles vary by gender?

Descriptive Statistics of Price by Gender:

	count	mean	std	min	25%	50%	75%	max
Gender								
Female	329.0	47705.167173	11244.836378	20000.0	38000.0	49000.0	55000.0	69000.0
Male	1199.0	32817.347790	12299.239195	18000.0	24000.0	29000.0	37500.0	70000.0
NOT Known	53.0	23339.622642	10308.591257	18000.0	18000.0	18000.0	19000.0	50000.0



Based on the previous analysis, here's how the amount spent on purchasing automobiles varies by gender:

Observations:

Female:

Mean Price: Females tend to purchase significantly more expensive cars on average, with a mean price of approximately $47,705.16.

Median Price: The median price for cars purchased by females is $49,000, indicating that half of their purchases are above this amount.

Distribution: Their purchases span a range from 25,000to 69,000. The distribution appears relatively symmetric around the median or slightly left-skewed, indicating a concentration of purchases at higher price points within their range.

Male:

Mean Price: Males, who constitute a much larger group, purchase cars with a lower average price of approximately $32,416.18.

Median Price: The median price for cars purchased by males is $29,000.

Distribution: Their purchases range from 18,000to 70,000. The distribution is right-skewed (mean > median), suggesting a tail extending to higher prices, implying some males purchase very expensive cars, but the majority are concentrated at lower price points compared to females.

NOT Known:

Mean Price: The 'NOT Known' category shows the lowest average expenditure at approximately $26,320.75.

Median Price: The median price for this group is $25,000. Distribution: Their price range is from 18,000to 47,000.

Key Takeaways:

Gender-based Spending: Females, despite being a smaller group in the dataset, generally purchase automobiles that are considerably more expensive than those purchased by males and the 'NOT Known' category, as indicated by both mean and median prices.

Price Range Overlap: While average spending differs, both males and females purchase across a wide range of prices, with maximum purchases reaching similar levels (69,000 for females, 70,000 for males). However, the central tendency of their spending is distinct.

5. How much money was spent on purchasing automobiles by individuals who took a personal loan?

To determine the total amount spent on automobiles by individuals who took a personal loan, calculate the sum of 'Price' for all customers who have a 'Personal_loan'. calculate the total money spent by individuals who took a personal loan.

The total money spent on automobiles by individuals who took a personal loan is $27,290,000.00.

6. How does having a working partner influence the purchase of higher-priced cars?

Based on the bivariate analysis, here's how having a working partner influences the purchase of higher-priced cars:

Higher Total Income: Individuals with a working partner generally have a higher 'Total_salary'. This increased household income allows them to afford and consequently purchase more expensive cars. Tendency Towards More Expensive Cars: The analysis showed that individuals with working partners tend to purchase slightly more expensive cars on average. This aligns with the understanding that greater financial capacity leads to higher-value purchases. Impact on Car Type: While the general preference for Sedans and Hatchbacks remains, the higher 'Total_salary' associated with a working partner means these households are more likely to be able to afford and therefore potentially choose higher-end models within any car make, or even transition to premium segments like SUVs if other demographic factors align. The Partner_salary itself shows a moderate positive correlation with 'Price' (0.42), further supporting this. In essence, having a working partner contributes to a higher combined household income, which acts as a significant enabler for the purchase of more expensive automobiles.

12. Key Recommendations & Insights

12.1. Data Analysis Key Findings

- **Strong Predictors of Car Price**: The individual's 'Salary' (0.83), 'Age' (0.79), and 'Total_salary' (0.76) exhibit strong positive correlations with the car's 'Price', indicating that higher income and older age are associated with more expensive vehicle

purchases.

- **Car Make Price Hierarchy**: 'SUVs' are positioned as premium vehicles with the highest median price and price range, while 'Sedans' fall in a moderate price bracket, and 'Hatchbacks' are the most affordable option.
- **Income Influences Car Type Choice**: Across all demographic categories analyzed, individuals purchasing 'SUVs' consistently demonstrate the highest 'Total_salary' distributions, followed by 'Sedan' buyers, and then 'Hatchback' buyers, reinforcing the link between higher income and more expensive car types.
- **Demographic Impact on Purchasing Power**: 'Business' professionals, 'Post Graduates', and 'Married' individuals generally show higher 'Total_salary' distributions and tend to purchase more expensive cars, especially 'SUVs', compared to their respective counterparts ('Salaried', 'Graduates', 'Single').
- **Consistent Car Make Preferences**: Despite 'SUVs' being associated with higher incomes, 'Sedan' and 'Hatchback' models are consistently preferred by a wider range of customers across various demographic and financial segments (Gender, Profession, Marital Status, Education, Loan Status, Partner Working).

12.2. Next Steps

- **Implement Income-Based Tiered Marketing**: Tailor marketing campaigns based on income levels and age, promoting premium 'SUV' models to high-income, older demographics, 'Sedans' to mid-income segments, and 'Hatchbacks' to budget-conscious, younger buyers.
- **Strategic Positioning for SUVs**: Given the general preference for 'Sedans' and 'Hatchbacks', 'SUV' marketing should be highly targeted towards higher-income segments like 'Business' professionals, 'Post Graduates', and 'Married' individuals, highlighting features that appeal to their specific needs and aspirations (e.g., status, family utility, advanced technology).

12.3. Actionable Insights & Recommendations:

1. **Which numerical variables have strong linear relationships with 'Price'?** 'Salary' (0.83), 'Age' (0.79), and 'Total_salary' (0.76) show strong positive correlations with 'Price'. 'Partner_salary' has a moderate positive correlation (0.42).
2. **Do certain car types ('Sedan', 'Hatchback', 'SUV') associate with distinct price ranges?** Yes, 'SUV's generally have the highest price range and median price, followed by 'Sedan's in the moderate range, and 'Hatchback's as the most budget-friendly option with the lowest price range and median price.
3. **Do specific customer segments tend to buy more expensive or cheaper cars?**
    - 'Business' professionals and 'Married' individuals tend to purchase more expensive cars compared to 'Salaried' and 'Single' individuals, respectively.
    - 'Post Graduates' tend to buy more expensive cars than 'Graduates'.
    - Males, on average, tend to purchase slightly more expensive cars than females. Females, despite being a smaller group in the dataset, generally purchase automobiles that are considerably more expensive than those purchased by males
    - Individuals with working partners also tend to purchase slightly more expensive cars due to higher combined incomes.
4. **Are there demographic preferences for different car types?** Yes, 'Sedan' and 'Hatchback' are consistently more popular than 'SUV' across almost all demographic and financial categories, despite 'SUV's being associated with higher income.